

ASSIGNMENT 2 REPORT

Gait Phase and EMG Analysis using DWT Denoising and CWT to Acquire Muscle-Activation Frequency Range



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CHAPTER I. FUNDAMENTAL THEORY

1.1. EMG Signal

The Electromyography (EMG) is a technique used to evaluate and record the electrical activity produced by skeletal muscles. The resulting signal, known as an electromyogram, detects the electric potential generated by muscle cells when these cells are electrically or neurologically activated. In gait analysis, Surface EMG (sEMG) is commonly used to assess the activation patterns of lower limb muscles. This study focuses on two specific muscles: the Gastrocnemius Lateralis (GL) and the Vastus Lateralis (VL). The Gastrocnemius Lateralis, located in the posterior calf, primarily functions in plantarflexion of the foot at the ankle and flexing the leg at the knee joint, playing a crucial role during the push-off phase of walking. The Vastus Lateralis, part of the quadriceps femoris group in the thigh, extends the knee and provides stability, particularly during the loading response phase of the gait cycle.

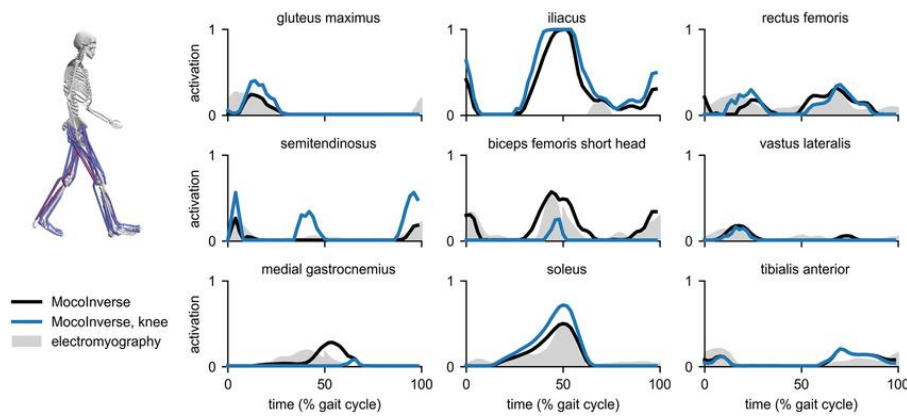


Figure 1.1. An example of EMG Signal after Preprocessing

1.2. Gait Signal

Human locomotion is a cyclic activity characterized by repetitive patterns of limb movement, commonly referred to as the gait cycle. A standard gait cycle is defined from the initial contact of one foot to the subsequent contact of the same foot. To accurately capture the temporal dynamics of this cycle, the Foot Switch signal serves as the primary reference, typically acquired using pressure sensors (basography) or Force Sensing Resistors (FSR) placed under the heel and toe. This signal provides a distinct representation of the foot-to-ground contact states, where specific amplitude transitions correspond to key gait events. Specifically, a sharp rise or high amplitude in the Foot Switch signal indicates the initial contact known as Heel Strike (HS), marking the beginning of the Stance Phase where the limb bears weight. Conversely, a drop in signal amplitude or a return to baseline indicates Toe Off (TO), marking the end of the Stance Phase and the onset of the Swing Phase where the foot clears the ground. The precise detection of these rising and falling edges within the Foot Switch signal is fundamental for segmenting continuous data into individual cycles and synchronizing the corresponding electromyographic signals.

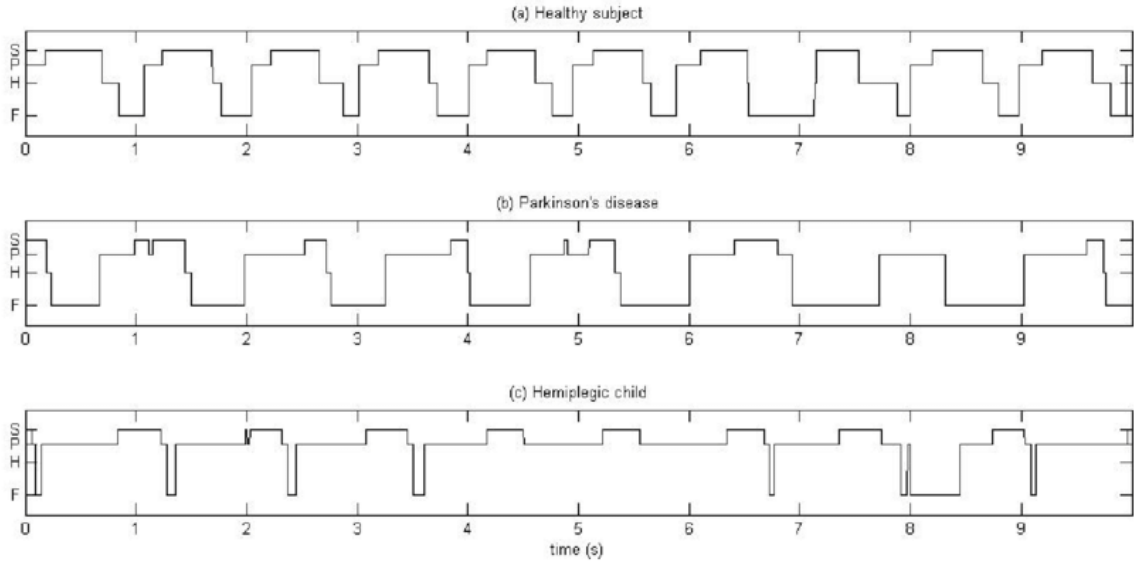


Figure 1.2. An example of Foot Switch Signal and Its Disease Condition

1.3. Segmentation on Gait to EMG

Segmentation is the process of partitioning continuous signals into discrete cycles to facilitate comparative analysis. While gait cycles are conventionally defined from Heel Strike to Heel Strike, specific analytical approaches may utilize a Toe-Off to Toe-Off segmentation strategy. This segmentation emphasizes the swing phase initiation and the subsequent stance phase, providing a specific window to observe muscle recruitment patterns required for limb advancement and subsequent weight acceptance. By isolating the signal between two consecutive Toe Off events (TO_i to TO_{i+1}), the system ensures that the analysis focuses on a complete functional unit of movement, allowing for the precise temporal localization of muscle activation within the normalized gait cycle.

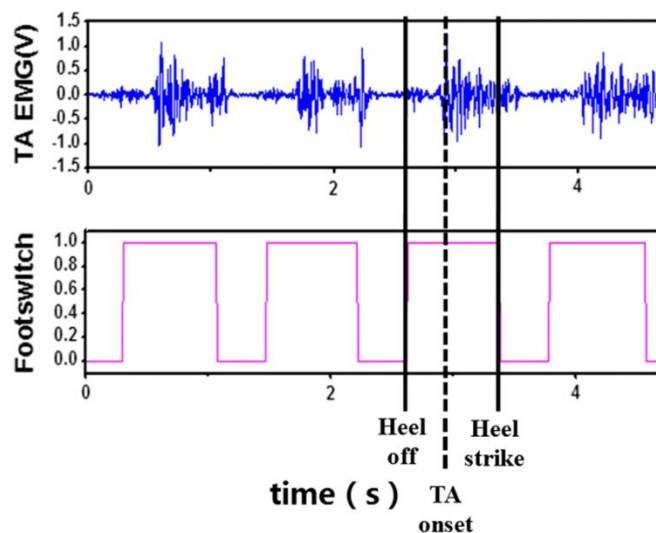


Figure 1.3. Example of A Segmentation between Foot Switch to EMG Signal

1.4. Butterworth 2nd Order BPF

Raw EMG signals often contain noise and motion artifacts that must be removed to extract the true muscle activity. A Band-Pass Filter is applied to attenuate frequencies outside the physiological range of muscle activity (typically 20-450 Hz). The Butterworth filter is

$$y[n] = b_0x[n] + b_1x[n-1] + b_2x[n-2] - a_1y[n-1] - a_2y[n-2]$$
 where $x[n]$ is the input signal, $y[n]$ is the filtered output, and a, b are the filter coefficients derived from the desired cutoff frequencies and sampling rate.

where $x[n]$ is the input signal, $y[n]$ is the filtered output, and a, b are the filter coefficients derived from the desired cutoff frequencies and sampling rate.

Figure 1.4. LPF and HPF Coefficients for Biquad Method from Robert Bristow-Johnson
Audio EQ Cookbook

The Discrete Wavelet Transform provides a time-scale representation of a signal, making it highly effective for denoising non-stationary signals like EMG. The DWT decomposes the signal into approximation coefficients (cA) and detail coefficients (cD) using a series of high-pass (g) and low-pass (h) filters, followed by downsampling. This process is governed by the Mallat algorithm. For EMG denoising, the Daubechies 4 (db4) mother wavelet is frequently selected due to its shape similarity to motor unit action potentials. The decomposition equations at level j are:

$$cA_j[k] = \sum_n x[n]h[2k-n], cD_j[k] = \sum_n x[n]g[2k-n]$$

Noise is typically reduced by applying a soft thresholding technique to the detail coefficients before reconstructing the signal via the Inverse DWT (IDWT).

1.6. Short Time Fourier Transform (STFT)

The Short-Time Fourier Transform is a technique used to determine the sinusoidal frequency and phase content of local sections of a signal as it changes over time. It addresses the limitation of the standard Fourier Transform, which loses temporal information. The STFT works by dividing the signal into short segments of equal length using a sliding window function $w[n]$ and computing the Fourier Transform for each segment. The mathematical representation of the STFT is:

$$X(m, \omega) = \sum_{n=-\infty}^{\infty} x[n]w[n-m]e^{-j\omega n}$$

where $x[n]$ is the signal and $w[n-m]$ is the window function shifted by m . This results in a time-frequency distribution (spectrogram) that visualizes how the spectral content of the EMG evolves during the gait cycle.

1.7. Window Method

In spectral analysis techniques such as STFT, window functions are applied to signal segments to mitigate spectral leakage caused by discontinuities at the boundaries. This method utilizes several specific window types, each defined mathematically for $0 \leq n \leq N-1$, where N represents the window length. The Rectangular window is the simplest form, defined as $w[n] = 1$, providing the best frequency resolution but poor dynamic range. The Triangular window reduces discontinuities with a linear taper:

$$w[n] = 1 - \left| \frac{n - \frac{N-1}{2}}{\frac{N-1}{2}} \right|$$

The Hanning window uses a cosine-squared bell shape to smooth the signal:

$$w[n] = 0.5 - 0.5\cos\left(\frac{2\pi n}{N-1}\right)$$

Similarly, the Hamming window offers specific coefficients to cancel the first side lobe:

$$w[n] = 0.54 - 0.46\cos\left(\frac{2\pi n}{N-1}\right)$$

The Blackman window provides greater attenuation using a higher-order cosine term:

$$w[n] = 0.42 - 0.5\cos\left(\frac{2\pi n}{N-1}\right) + 0.08\cos\left(\frac{4\pi n}{N-1}\right)$$

Finally, the Kaiser window allows for adjustable side-lobe attenuation via the parameter β , utilizing the zero-th order modified Bessel function of the first kind, I_0 :

$$w[n] = \frac{I_0\left(\beta\sqrt{1 - \left(\frac{2n}{N-1} - 1\right)^2}\right)}{I_0(\beta)}$$

1.8. Continuous Wavelet Transform (CWT)

The Continuous Wavelet Transform analyzes the time-frequency characteristics of the EMG signal by correlating it with scaled and translated versions of a mother wavelet $\psi(t)$. The transformation is mathematically governed by:

$$CWT(a, b) = \frac{1}{\sqrt{a}} \int_{-\infty}^{\infty} x(t) \psi^*\left(\frac{t-b}{a}\right) dt$$

This study employs two distinct mother wavelets. The first is the Morlet wavelet, which is analytically defined as a complex exponential modulated by a Gaussian window:

$$\psi(t) = \pi^{-0.25} e^{i\omega_0 t} e^{-t^2/2}$$

where ω_0 is the central frequency parameter (typically set to 6 to satisfy admissibility). The second is the Daubechies 4 (db4) wavelet. Unlike Morlet, db4 lacks a simple closed-form analytical expression and is instead constructed via the Cascade Algorithm using a scaling function $\phi(t)$ and a wavelet function $\psi(t)$. These are derived from the filter coefficients $h[n]$ (low-pass) and $g[n]$ (high-pass) through the dilation equations:

$$\phi(t) = \sqrt{2} \sum_n h[n] \phi(2t - n) \text{ and } \psi(t) = \sqrt{2} \sum_n g[n] \phi(2t - n)$$

where $g[n] = (-1)^n h[1 - n]$. The db4 wavelet is asymmetric and orthogonal, making it particularly effective for detecting transient spikes and discontinuities characteristic of motor unit firing in EMG signals.

1.9. Thresholding

To quantify muscle activity timing, the continuous energy distribution from the CWT must be converted into binary activation states. This is achieved through a thresholding process. First, a 1D energy envelope is obtained by integrating the CWT scalogram across frequencies. An adaptive threshold is then established, typically defined as a percentage (e.g., 1%) of the peak energy within the cycle:

$$Threshold = \alpha \cdot \max(E(t))$$

The Onset timing is defined as the point where the energy profile rises above this threshold, and the Offset is where it falls below. To ensure physiological relevance, temporal constraints are applied: brief gaps between activations like, < 30 ms are merged, and activation durations shorter than a minimum physiological limit for example, < 30 ms are discarded as noise.

CHAPTER II. RESULT AND ANALYSIS

2.1. Problems Statement

Step:

- Load and plot raw data → Channel 7, 10, 13
- Segment data based on Foot Switch (FS) data
- Filter using BPF
- Denoise data using DWT
- STFT
- CWT
- Determine the onset and offset (1% TH)

2.2. Code Explanation

This section will explain the practical implementation of the signal processing pipeline designed to analyze non-stationary EMG and gait signals. Developed using the Python programming language and the PyQt6 framework for the Graphical User Interface (GUI), the system integrates distinct modules ranging from raw data acquisition to advanced time-frequency analysis and threshold-based event detection. A key characteristic of this implementation is the manual derivation and coding of mathematical algorithm, including the Butterworth filter, Discrete Wavelet Transform via Mallat's algorithm, and Continuous Wavelet Transform. This approach ensures a fundamental verification of the theoretical concepts discussed in Chapter I. The following sections provide a detailed explanation of the code structure, the mathematical formulations applied within each script, and an analysis of the resulting waveforms and detection accuracy.

For brevity purposes, the code lines snippet will not be showed fully, because each modules have many lines of code, especially GUI script. So, in the below explanation I will only showed the main class or main function on each script.

a. Data Loading and Visualization (*Load_and_Plot_Raw_Data.py*)

The initial phase involves loading physiological signal data from Physionet records. This script utilizes the wfdb library to handle the specific data formats used in physiological databases.

```
1. import wfdb
2. # ... (and other imports)
3.
4. TARGET_CHANNELS = {
5.     'FS': ['baso RT FOOT', 'baso RT FOOT'],
6.     'GL': ['semg RT LAT.G', 'semg RT LAT.G'],
7.     'VL': ['semg RT LAT.V', 'semg RT LAT.V']
8. }
9.
10. def load_raw_data(record_path):
```

```

11. record = wfdb.rdrecord(record_path)
12. # ... (channel searching logic)
13.
14. # Extract Signals
15. raw_fs = record.p_signal[:, idx_fs]
16. raw_gl = record.p_signal[:, idx_gl]
17. raw_vl = record.p_signal[:, idx_vl]
18.
19. # Create time array
20. total_samples = len(raw_fs)
21. time = np.arange(total_samples) / fs
22.
23. return { ... }

```

The `load_raw_data` function is the core component here. It reads the header (.hea) and data (.dat) files using `wfdb.rdrecord`. A critical feature is the dynamic channel search mechanism implemented via `get_channel_index` and the `TARGET_CHANNELS` dictionary. This ensures robustness against slight variations in channel naming conventions across different records like trailing spaces. The function extracts the signals for the Foot Switch (FS), Gastrocnemius Lateralis (GL), and Vastus Lateralis (VL) from the `p_signal` array, which contains the calibrated physical values (e.g., voltage in mV) rather than raw digital integers. A time vector t is also generated based on the sampling frequency (f_s) and the total number of samples (N):

$$t[n] = \frac{n}{f_s}, \text{ for } n = 0, 1, \dots, N - 1$$

b. *Gait Cycle Segmentation (Segmentation_Foot_Switch.py)*

This script is for partitioning the continuous EMG signals into discrete gait cycles based on the Foot Switch signal. The segmentation logic is crucial for analyzing muscle activation within consistent functional windows.

```

1. def detect_events_manual(fs_signal, fs, min_dist_sec=0.4):
2.     # 1. Dynamic Thresholding
3.     t_min = np.min(fs_signal)
4.     t_max = np.max(fs_signal)
5.     threshold = (t_max + t_min) / 2
6.
7.     # 2. Binarization
8.     binary_signal = (np.abs(fs_signal) > np.abs(threshold)).astype(int)
9.
10.    # 3. Edge Detection
11.    edges = np.diff(binary_signal)
12.    raw_hs = np.where(edges == 1)[0] # Rising Edge (Heel Strike)
13.    raw_to = np.where(edges == -1)[0] # Falling Edge (Toe Off)
14.
15.    # 4. Manual Debouncing
16.    # ... (debounce logic)

```


The detection algorithm operates on the Foot Switch signal, which exhibits a high amplitude during ground contact (Stance phase) and a low amplitude during the swing phase.

1. A threshold is calculated adaptively as the midpoint between the signal's minimum and maximum values: $T_{hresh} = \frac{\min(x) + \max(x)}{2}$.
2. The analog signal $x[n]$ is converted into a binary state signal $b[n]$:
$$b[n] = \begin{cases} 1 & \text{if } |x[n]| > |T_{hresh}| \\ 0 & \text{otherwise} \end{cases}$$
3. Transitions are identified using the first-order difference: $d[n] = b[n + 1] - b[n]$. A result of +1 indicates a rising edge (Heel Strike), while -1 indicates a falling edge (Toe Off).
4. To prevent false detections due to signal noise or mechanical bounce, a refractory period (`min_dist_sec`) is applied. Events detected within this window after a valid event are ignored.

```

1. def segment_data(data_dict):
2.     # ... (event detection call)
3.
4.     # Cycle = Toe Off [i] -> Toe Off [i+1]
5.     for i in range(len(toe_offs) - 1):
6.         start_idx = toe_offs[i]
7.         end_idx = toe_offs[i+1]
8.
9.         # Find Heel Strike within this cycle
10.        cycle_hs_candidates = heel_strikes[(heel_strikes > start_idx) &
11.        (heel_strikes < end_idx)]
11.        # ... (slicing and storage)

```

The `segment_data` function is for the slicing. It defines a single gait cycle as the interval between two consecutive Toe Off events (TO_i to TO_{i+1}). This captures the full Swing Phase followed by the Stance Phase. Within each cycle, the script specifically searches for the internal Heel Strike event (HS), which marks the transition from Swing to Stance. The resulting segments, containing synchronized GL, VL, and FS data, are stored in a list of dictionaries for subsequent processing stages.

c. *Band-Pass Filtering (Filtering_BPF.py)*

After segmentation, the raw EMG signals often contain motion artifacts (low-frequency noise) and high-frequency interference that must be attenuated. This script implements a digital 2nd-order Band-Pass Filter (BPF) with a passband of 20–450 Hz. The implementation offers two distinct methods: a standard BPF derived from an analog prototype via the Bilinear Transform, and a cascade of High-Pass and Low-Pass Biquad filters based on the Robert Bristow-Johnson (RBJ) Audio EQ Cookbook.

```

1. class ManualFilterDesigner:
2.     def compute_standard_bpf_coeffs(self, f_low, f_high):
3.         # 1. Pre-warping frequencies
4.         wd_low = 2 * self.fs * np.tan((2 * np.pi * f_low) / (2 * self.fs))
5.         wd_high = 2 * self.fs * np.tan((2 * np.pi * f_high) / (2 * self.fs))
6.

```

```

7.      # 2. Analog Parameters (Center Freq & Bandwidth)
8.      w0_sq = wd_low * wd_high
9.      bw = wd_high - wd_low
10.
11.     # 3. Bilinear Substitution Constants (s -> z)
12.     C = 2 * self.fs
13.     # ... (algebraic expansion for coefficients a and b)
14.     return b, a

```

The core of the standard method lies in `compute_standard_bpf_coeffs`. It starts by pre-warping the desired analog cutoff frequencies (f_{low}, f_{high}) to the digital domain to correct for the non-linear mapping of the Bilinear Transform near the Nyquist frequency. The analog center frequency squared (ω_0^2) and bandwidth (BW) are then calculated. By substituting the s-domain transfer function $H(s) = \frac{BW \cdot s}{s^2 + BW \cdot s + \omega_0^2}$ with the

bilinear relation $ss = \frac{2}{T} \frac{1-z^{-1}}{1+z^{-1}}$, the script algebraically derives the digital coefficients b (numerator) and a (denominator) for the difference equation. This manual derivation ensures the filter characteristics precisely match the theoretical analog prototype without relying on black-box filter design functions. The filtering process itself uses `filtfilt_manual`, a zero-phase filtering technique that processes the signal forward and backward, effectively doubling the filter order but eliminating phase distortion, which is crucial for preserving the timing of muscle activation onsets.

d. *Wavelet Denoising (Denoising_DWT_EMG.py)*

Following the band-pass filter, the signal may still contain non-stationary noise components. This script applies Discrete Wavelet Transform (DWT) denoising using the Mallat algorithm (Filter Bank approach) with the Daubechies 4 (db4) mother wavelet. The process involves decomposing the signal into multiple levels of approximation and detail coefficients, thresholding the noise, and then reconstructing the clean signal.

```

1. class ManualDWT:
2.     def dwt_decomposition(self, signal, levels=8):
3.         # ...
4.         for i in range(levels):
5.             # Convolve using 'valid' mode after padding
6.             cA = np.convolve(padded, self.ld_coeffs, mode='valid') # Low-pass
7.             cD = np.convolve(padded, self.hd_coeffs, mode='valid') # High-pass
8.
9.             # Downsampling (Decimation)
10.            cA_down = self.downsample(cA)
11.            cD_down = self.downsample(cD)
12.            # ...

```

The `ManualDWT` class manually defines the decomposition low-pass (L_D) filter coefficients for the db4 wavelet. The high-pass coefficients (H_D) are derived using the Quadrature Mirror Filter relationship: $H_D[n] = (-1)^n L_D[N - 1 - n]$. The `dwt_decomposition` function iteratively applies these filters. At each level j , the signal (starting with the input approximation) is convolved with both filters and then

downsampled by a factor of 2 (keeping every even sample). The resulting detail coefficients (cD_j) capture high-frequency information (often noise), while the approximation coefficients (cA_j) are passed to the next level.

For denoising, the script uses a soft thresholding technique. The noise level (σ) is estimated from the first level of detail coefficients (cD_1) using the Median Absolute Deviation (MAD): $\sigma = \text{median}(|cD_1|)/0.6745$. A universal threshold $T = \sigma\sqrt{2\ln(N)}$ is then calculated. The `soft_thresholding` function shrinks coefficients towards zero: values smaller than T become zero, while larger values are reduced by T . Finally, the `idwt_reconstruction` function reverses the process using upsampling (inserting zeros) and convolution with reconstruction filters to synthesize the denoised EMG signal. A critical addition in this script is the manual windowing class (`ManualWindow`), which allows the user to apply various window functions (Hanning, Hamming, etc.) to the signal segment before processing, further reducing spectral leakage at the boundaries.

e. *Short-Time Fourier Transform (STFT_EMG.py)*

The STFT is implemented to analyze the non-stationary characteristics of the EMG signal by examining its frequency content over short, overlapping time windows. This script provides a manual implementation of the STFT algorithm, incorporating a custom Cooley-Tukey Fast Fourier Transform (FFT) and a variety of window functions.

```

1. class ManualSTFT:
2.     def compute(self, signal, fs):
3.         # ... (segmentation logic)
4.         for i in range(n_segments):
5.             # Apply Window
6.             windowed_segment = segment * window
7.
8.             # Compute Manual FFT
9.             spectrum, n_fft = self.fft_solver.compute_fft(windowed_segment)
10.
11.            # Magnitude Normalization
12.            magnitude = np.abs(spectrum[:half_n]) / n_fft
13.            stft_matrix.append(magnitude)
14.            # ...

```

The `ManualSTFT` class first generates the chosen window function (e.g., Hanning, Hamming) using the `ManualWindow` class to minimize spectral leakage. The core computation involves segmenting the signal into overlapping frames defined by `window_size` and `overlap`. For each frame, the script applies the window and computes the spectrum using `compute_fft`, which utilizes a recursive Radix-2 FFT algorithm. The magnitude of the positive frequency components is then calculated: $|X[k]| = \sqrt{\text{Re}(X[k])^2 + \text{Im}(X[k])^2}$. Finally, the resulting spectral frames are stacked to form a time-frequency matrix (spectrogram). For visualization, the time axis is normalized to the percentage of the gait cycle (0-100%), allowing for standardized comparison across different strides.

f. *Continuous Wavelet Transform (CWT_EMG.py)*

The CWT serves as the primary tool for detecting muscle activation timing in this study due to its superior time-frequency resolution trade-off compared to STFT. This script implements the CWT manually by convolving the EMG signal with scaled and translated versions of a mother wavelet. The system supports two mother wavelets: the analytical Morlet wavelet and the numerical Daubechies 4 (db4).

```
1. class ManualCWT:
2.     def compute(self, signal):
3.         # ...
4.         for i, scale in enumerate(self.scales):
5.             # 1. Generate Mother Wavelet at current scale
6.             psi = self._morlet_wavelet(scale) # or _db4_wavelet
7.
8.             # 2. Perform Convolution (Signal * Wavelet)
9.             coeffs = np.convolve(signal, psi, mode='same')
10.            cwt_matrix[i, :] = coeffs
11.
12.            # Calculate Scalogram (Energy Density)
13.            energy_density = np.abs(cwt_matrix)**2
14.        return energy_density, self.freqs
```

The scale parameter a is inversely proportional to frequency, determined by the relationship $a = \frac{f_c f_s}{f}$, where f_c is the center frequency of the wavelet. For the Morlet wavelet, the shape is generated using the analytical formula $\psi(t) = \pi^{-0.25} e^{i\omega_0 t} e^{-t^2/2}$, which provides excellent frequency localization. For the db4 wavelet, which lacks a closed-form expression, the script employs the Cascade Algorithm (`_generate_db4_prototype`) to iteratively construct the wavelet shape from filter coefficients, followed by interpolation to match the required scale. The core transformation is performed via `np.convolve`, calculating the inner product between the signal and the wavelet at each scale. The final output is the Scalogram, defined as the squared magnitude of the CWT coefficients ($|W(a, b)|^2$), representing the energy density of the signal in the time-scale plane. This energy distribution forms the basis for the subsequent threshold-based onset/offset detection.

g. *Muscle Activation Thresholding (Threshold.py)*

The final step in processing the EMG signal is to convert the continuous time-frequency energy distribution into discrete binary states of muscle activity (Active vs. Inactive). This script implements an adaptive thresholding algorithm combined with physiological constraints to robustly detect the Onset and Offset timings.

```
1. def detect_bursts(energy_profile, fs, threshold_ratio=0.01):
2.     # 1. Absolute Thresholding
3.     threshold_val = threshold_ratio * np.max(energy_profile)
4.     is_active = (energy_profile > threshold_val).astype(int)
5.
6.     # 2. Edge Detection
```

```

7.    diff = np.diff(np.pad(is_active, (1, 1), 'constant'))
8.    onsets = np.where(diff == 1)[0]
9.    offsets = np.where(diff == -1)[0]
10.
11.   # 3. Physiological Constraints (Debouncing)
12.   # Merging gaps < 30ms and discarding bursts < 30ms
13.   # ... (loop implementation)

```

First, the 2D scalogram $|W(a, b)|^2$ obtained from the CWT is collapsed into a 1D energy profile $E(t)$ by summing the energy across all frequency scales (marginal integration). An adaptive threshold T is then calculated as 1% of the peak energy within the cycle: $T = 0.01 \cdot \max(E(t))$. The signal is binarized based on this threshold. Edge detection is performed using the discrete difference operator to identify potential Onset ($0 \rightarrow 1$) and Offset ($1 \rightarrow 0$) points. To eliminate false positives caused by signal transients or noise, two critical physiological constraints are applied manually:

1. If the gap between two consecutive activations is less than 30 ms, they are merged into a single continuous burst.
2. If the duration of a detected burst is less than 30 ms, it is considered non-physiological noise and removed. This dual-constraint approach ensures that the final detected intervals represent significant, sustained muscle contractions.

The main.py script serves as the entry point (main execution point) of the software. This script is very simple because its task is only to initialize the application's event loop using QApplication and call the main PCGANalyzerGUI class to display the program window. This separation is done to separate the interface logic (GUI.py) from the program's execution logic, adhering to good programming standards.

h. Result Reporting and Metric Calculation (Result_Reporting.py)

Once the activation intervals are determined, this script quantifies the results into a readable report format. It extracts the absolute timestamps of the detected Onset and Offset events and converts them into relative metrics standard for gait analysis.

```

1. def detect_bursts(energy_profile, fs, threshold_ratio=0.01):
2.     # 1. Absolute Thresholding
3.     threshold_val = threshold_ratio * np.max(energy_profile)
4.     is_active = (energy_profile > threshold_val).astype(int)
5.
6.     # 2. Edge Detection
7.     diff = np.diff(np.pad(is_active, (1, 1), 'constant'))
8.     onsets = np.where(diff == 1)[0]
9.     offsets = np.where(diff == -1)[0]
10.
11.    # 3. Physiological Constraints (Debouncing)
12.    # Merging gaps < 30ms and discarding bursts < 30ms
13.    # ... (loop implementation)

```

The primary function, generate_cycle_report, takes a processed gait cycle segment as input. For each detected muscle burst (GL and VL), it calculates the timing relative to

the specific gait cycle duration. The Onset and Offset times are normalized to a percentage of the gait cycle (%GC) using the formula:

$$T_{\%GC} = \frac{t_{event} - t_{start}}{t_{end} - t_{start}} \times 100\%$$

where t_{start} is the Heel Strike initiating the cycle. Additionally, the absolute duration of each activation is calculated in milliseconds. This module formats these quantitative metrics into a structured text block, which is then displayed in the GUI alongside the visual plots, providing a comprehensive clinical summary of the muscle recruitment pattern for each step.

i. Graphical User Interface (GUI.py)

The GUI serves as the central controller for the entire analysis pipeline, providing an interactive platform for users to visualize each stage of signal processing. Developed using the PyQt6 framework, the GUI is structured into a tabbed layout, where each tab corresponds to a specific processing module/step that are Load Data, Segmentation, Filtering, Denoising, STFT, CWT, and Thresholding. This modular design allows for step-by-step verification of the algorithms. The following lines are summarized for brevity (because the GUI script are very long).

```
1. class MainWindow(QMainWindow):
2.     def __init__(self):
3.         # ...
4.         self.create_tab_load()
5.         self.create_tab_segmentation()
6.         # ... (other tabs)
```

Key features of the GUI implementation include:

1. A dedicated MplCanvas class integrates Matplotlib figures directly into the PyQt window. It is customized with a dark theme (dark background, white axes/labels) to enhance visual clarity for complex spectral plots.
2. The plotting logic is designed to handle user interactions dynamically. For instance, changing the cycle index via a QSpinBox instantly updates the zoomed views of the Foot Switch and EMG segments without reloading the entire dataset.
3. The GUI exposes critical algorithm parameters to the user, such as the window type for STFT, the mother wavelet for CWT (Morlet vs. db4), and the percentage threshold for activation detection. This flexibility allows for real-time tuning and comparative analysis.
4. To prevent visual clutter, the GUI employs QVBoxLayout and QScrollArea widgets. Advanced layouts, such as the split view in the Thresholding tab, combine graphical plots with a textual report panel (QTextEdit), enabling simultaneous qualitative and quantitative assessment.

j. Main Execution (main.py)

The main.py script acts as the entry point for the application. Its primary responsibility is to initialize the PyQt application instance, configure global settings, and launch the main window.

```
1. def main():
2.     # ...
3.     app = QApplication(sys.argv)
4.     app.setStyle("Fusion") # Consistent base style
5.
6.     # High DPI Scaling for modern displays
7.     if hasattr(Qt.ApplicationAttribute, 'AA_UseHighDpiPixmaps'):
8.         app.setAttribute(Qt.ApplicationAttribute.AA_UseHighDpiPixmaps)
9.
10.    window = MainWindow()
11.    window.showMaximized()
12.    sys.exit(app.exec())
```

This script ensures that the application environment is correctly set up before loading the heavy GUI components. It explicitly sets the application style to "Fusion" to provide a consistent look across different operating systems and enables High-DPI scaling attributes to ensure that the detailed signal plots remain sharp on high-resolution monitors. Error handling blocks are included to catch import errors (e.g., missing libraries) and initialization failures, providing informative feedback to the console for debugging purposes.

2.3. Result of the Program and Analysis

*Initial GUI

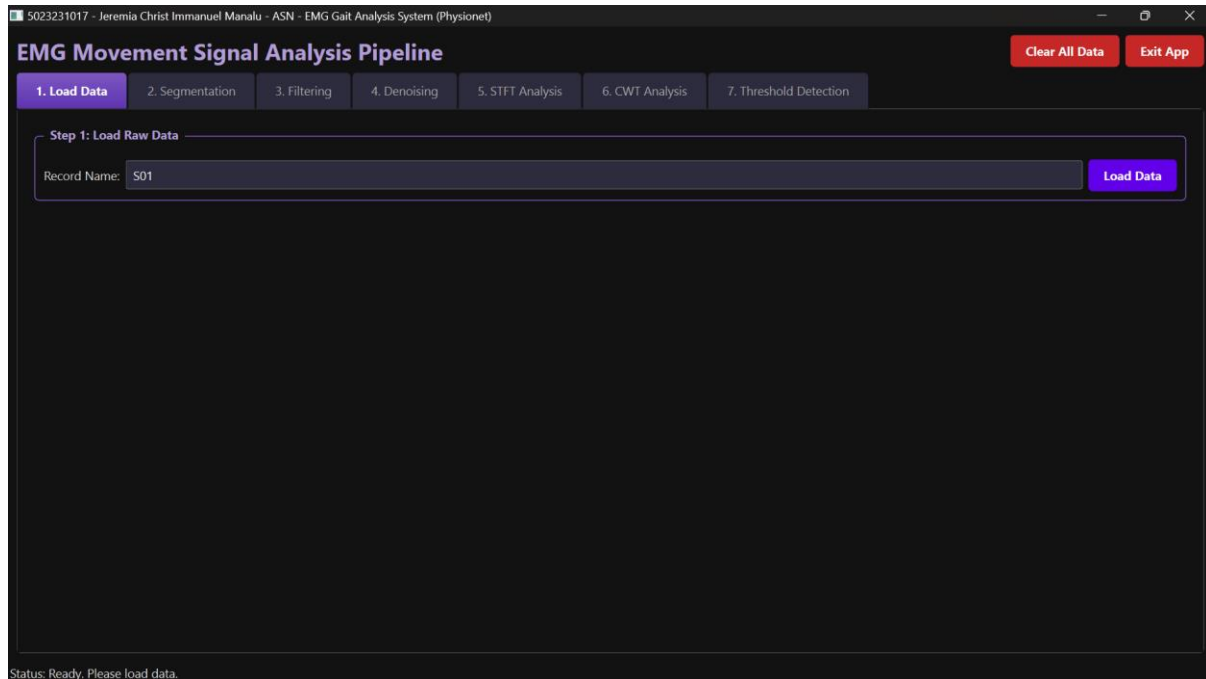


Figure 2.1. Initial GUI Components for the Program

The implementation of the signal processing algorithms discussed in the previous section culminates in a comprehensive desktop application. The Graphical User Interface (GUI), developed using the PyQt6 framework (as seen in **Figure 2.1**), is designed with a modern dark-themed aesthetic to enhance visual contrast for signal inspection. The interface is structured into a sequential pipeline consisting of seven distinct tabs: Load Data, Segmentation, Filtering, Denoising, STFT Analysis, CWT Analysis, and Threshold Detection. This tabbed architecture enforces a logical workflow, ensuring that the user validates the data at every stage of processing, from raw signal acquisition to the final determination of muscle activation timing. The initial view of the program, as shown in the application window, presents the data loading module where specific patient records can be retrieved for analysis.

2.3.1. Dataset Explanation

The data used in this section is S01.dat-S01.heg and S7.dat-S7.heg. The detail of the .heg file for S01.heg is as below:

1. S01 14 2000 611001
2. S01.dat 16 2.0686(-28928)/mV 0 0 -29127 -3860 0 baso LT FOOT
3. S01.dat 16 1350.8768(-29593)/mV 0 0 -32394 16595 0 gonio RT HIP
4. S01.dat 16 47.9561(-7230)/mV 0 0 -7422 16499 0 semg LT TIB.A
5. S01.dat 16 32.7088(-352)/mV 0 0 -303 5369 0 semg LT LAT.G
6. S01.dat 16 139.1338(765)/mV 0 0 626 -22151 0 semg LT REC.F
7. S01.dat 16 50.8591(954)/mV 0 0 1005 -14966 0 semg LT HAM
8. S01.dat 16 34.4181(-1807)/mV 0 0 -1773 -17085 0 semg LT LAT.V
9. S01.dat 16 2.1422(-30642)/mV 0 0 -30676 23102 0 baso RT FOOT

10. S01.dat 16 848.225(-28429)/mV 0 0 -29250 -3547 0 gonio RT KNEE
11. S01.dat 16 32.045(32)/mV 0 0 176 13306 0 semg RT TIB.A
12. S01.dat 16 43.9075(3875)/mV 0 0 4029 2298 0 semg RT LAT.G
13. S01.dat 16 71.1917(4467)/mV 0 0 4645 -16440 0 semg RT REC.F
14. S01.dat 16 113.7708(2674)/mV 0 0 3243 -12745 0 semg RT HAM
15. S01.dat 16 76.2443(-5623)/mV 0 0 -5242 -5957 0 semg RT LAT.V

The detail of the .hea file for S01.hea is as below:

1. S7 14 2000 618861
2. S7.dat 16 2.1557(-30560)/V 0 0 -30526 29804 0 baso LT FOOT
3. S7.dat 16 903.1576(-32517)/deg 0 0 -32642 -18709 0 gonio LT KNEE
4. S7.dat 16 56.8607(2061)/uV 0 0 1919 -16329 0 semg LT TIB.A
5. S7.dat 16 37.3189(-3639)/uV 0 0 -3602 3886 0 semg LT LAT.G
6. S7.dat 16 65.7622(-345)/uV 0 0 -477 -25339 0 semg LT REC.F
7. S7.dat 16 41.8601(-5327)/uV 0 0 -5453 2950 0 semg LT HAM
8. S7.dat 16 100.2018(2255)/uV 0 0 2205 26906 0 semg LT LAT.V
9. S7.dat 16 2.1591(-31005)/V 0 0 -31005 -18953 0 baso RT FOOT
10. S7.dat 16 975.6333(-28317)/deg 0 0 -28047 -645 0 gonio RT KNEE
11. S7.dat 16 74.5529(1976)/uV 0 0 1939 28880 0 semg RT TIB.A
12. S7.dat 16 48.9045(-293)/uV 0 0 -171 32247 0 semg RT LAT.G
13. S7.dat 16 186.1705(-2793)/uV 0 0 -2421 -3410 0 semg RT REC.F
14. S7.dat 16 53.5392(-910)/uV 0 0 -1017 7767 0 semg RT HAM
15. S7.dat 16 64.7549(486)/uV 0 0 777 19996 0 semg RT LAT.V

The system processes physiological data sourced from the Physionet database, specifically utilizing records the above **S01** and **S7** records. These datasets are characterized by a high sampling frequency of **2000 Hz**, which is essential for capturing the rapid transient features of motor unit action potentials and precise gait events. While the raw files (.hea and .dat) contain multi-channel recordings including kinematics (goniometry) and various muscle groups, this study specifically isolates three channels from the right lower limb for processing. These are the **Foot Switch signal** (baso RT FOOT), used as the ground truth for gait cycle segmentation, and the surface electromyography (sEMG) signals from the **Gastrocnemius Lateralis** (semg RT LAT.G) and **Vastus Lateralis** (semg RT LAT.V). The raw data amplitudes are calibrated in millivolts (mV), providing a direct physiological representation of muscle electrical activity before any digital filtering is applied.

2.3.2. Program Result for S01 Data

This section presents the results of the signal processing pipeline applied to the **S01** dataset obtained from Physionet. The S01 record contains surface electromyography (sEMG) and gait data sampled at 2000 Hz. The analysis proceeds sequentially through data loading, segmentation, filtering, time-frequency analysis, and threshold-based activation detection, demonstrating the efficacy of the implemented algorithms in extracting muscle activation patterns from non-stationary signals.

1. Load Data Tab

The initial stage involves visualizing the raw signals to verify data integrity. **Figure 2.2** displays three time-series plots extracted from the S01 record over a duration of approximately 300 seconds. The top graph shows the raw Foot Switch signal (Channel 7),

characterized by a square-wave-like pattern that toggles between high and low states, corresponding to the stance and swing phases of the gait cycle. The middle and bottom graphs display the raw sEMG signals for the GL (Channel 10) and VL (Channel 13), respectively. These signals exhibit the stochastic, noise-like properties typical of raw sEMG, with amplitudes fluctuating in the millivolt (mV) range. The high-density bursts in the EMG signals visually correlate with the periodic activity observed in the foot switch signal, indicating muscle recruitment during locomotion.

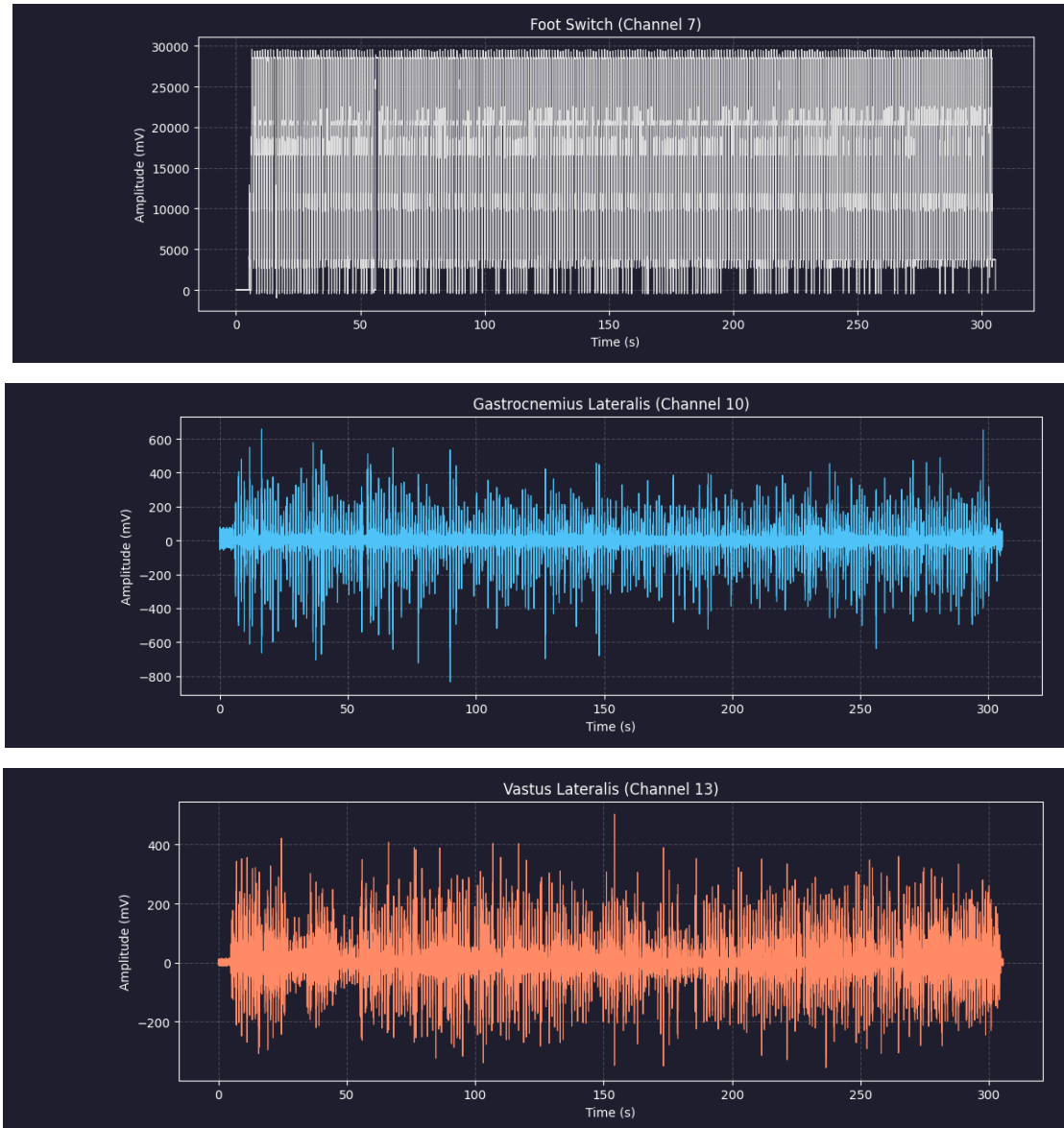


Figure 2.2. Three Graphs Result of the Load Data Tab for *S01 Data*

2. Segmentation Tab

Following data loading, the continuous signal is segmented into individual gait cycles. **Figure 2.3** illustrates the segmentation result for a specific cycle, Cycle 144. The top graph shows the segmented Foot Switch signal, where the vertical dashed lines mark the critical gait events: the Toe Off (start of the cycle, red dashed line), the Heel Strike (cyan dotted line), and the subsequent Toe Off (end of the cycle, purple dashed line). This segmentation defines a full gait cycle starting from the swing phase. The middle and bottom graphs show

the corresponding raw GL and VL signals isolated for this specific time window. By isolating Cycle 144, we can observe the specific raw muscle activity associated with one complete stride, ready for further processing.

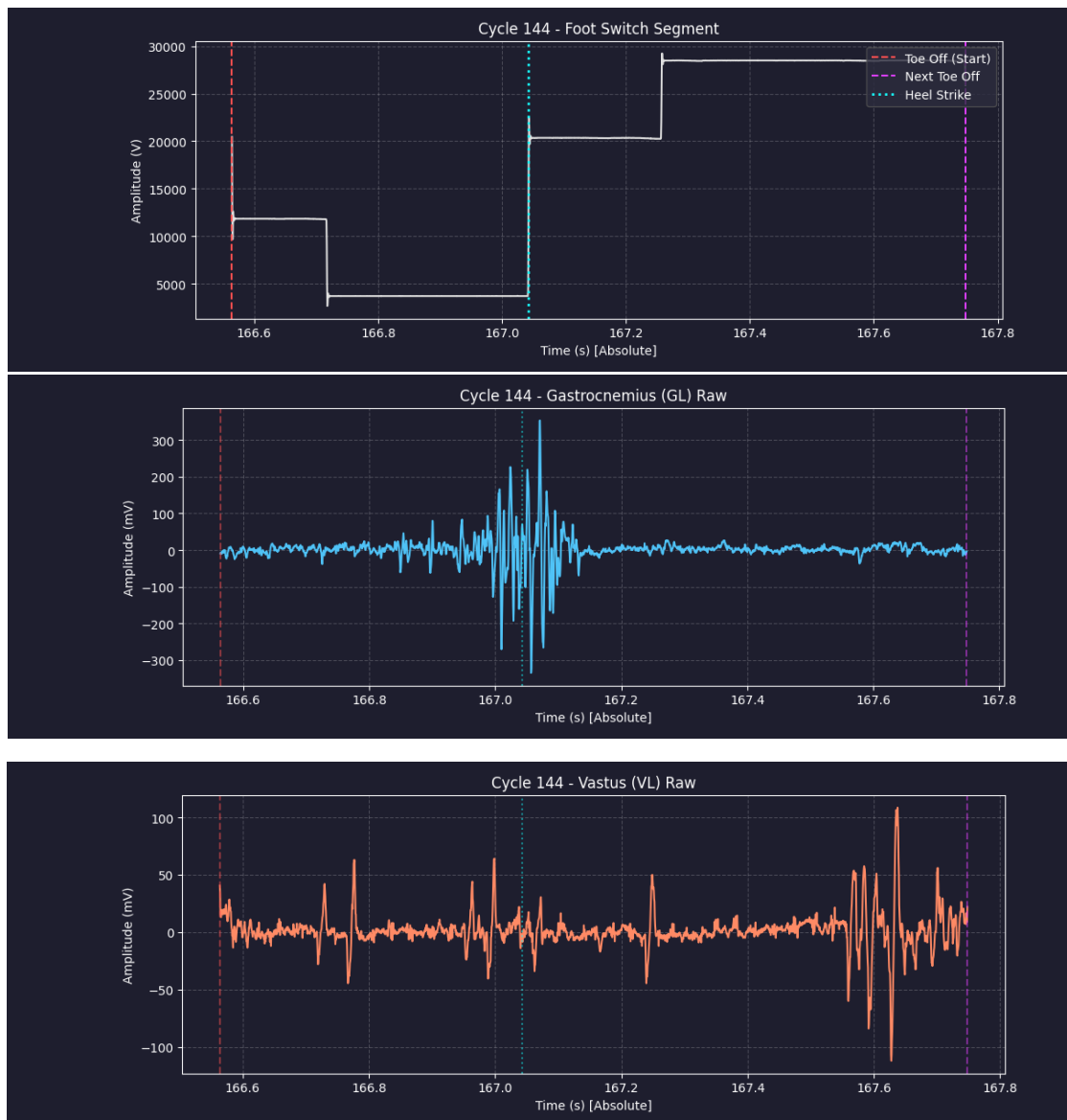


Figure 2.3. The Segmentation Result for GL and VL Signal on Cycle 144

3. Butterworth 2nd Order BPF Tab

To remove motion artifacts and high-frequency noise, a BPF is applied. **Figure 2.4** presents the frequency response comparison between the two implemented filtering methods: the Standard 2nd Order Butterworth BPF (cyan line) derived via the Bilinear Transform, and the RBJ Biquad Cascade method (magenta dashed line). The plot confirms that both methods achieve a similar passband profile, effectively attenuating frequencies below 20 Hz and above 450 Hz, though slight differences in the roll-off characteristics are visible.

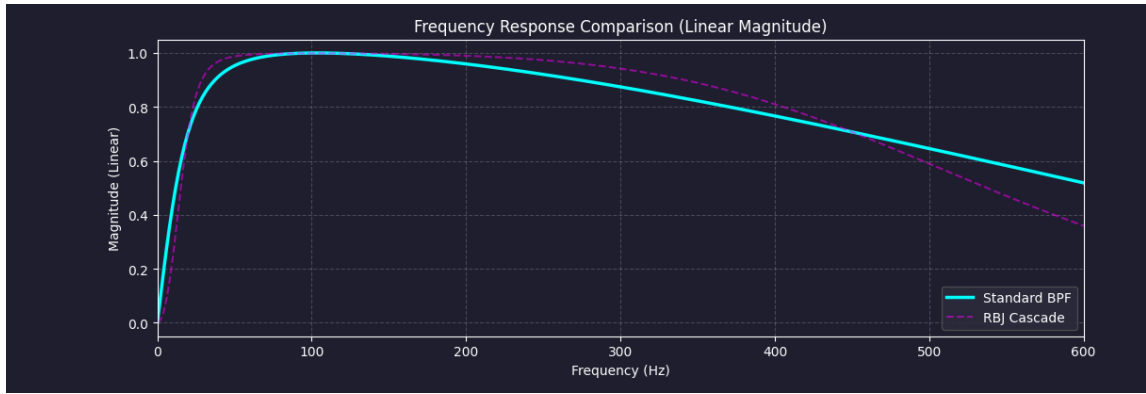


Figure 2.4. The Frequency Response Plot for Both Standard 2nd Order Butterworth BPF and with RBJ Biquad Cascade Method

Figure 2.5 subsequently shows the time-domain result of applying the RBJ Cascade BPF to the GL and VL signals of Cycle 144. Compared to the raw signals in **Figure 2.3**, the filtered signals (orange/blue traces overlaying the gray raw signal) appear centered around zero with reduced baseline wander, indicating effective removal of low-frequency artifacts while preserving the burst characteristics.

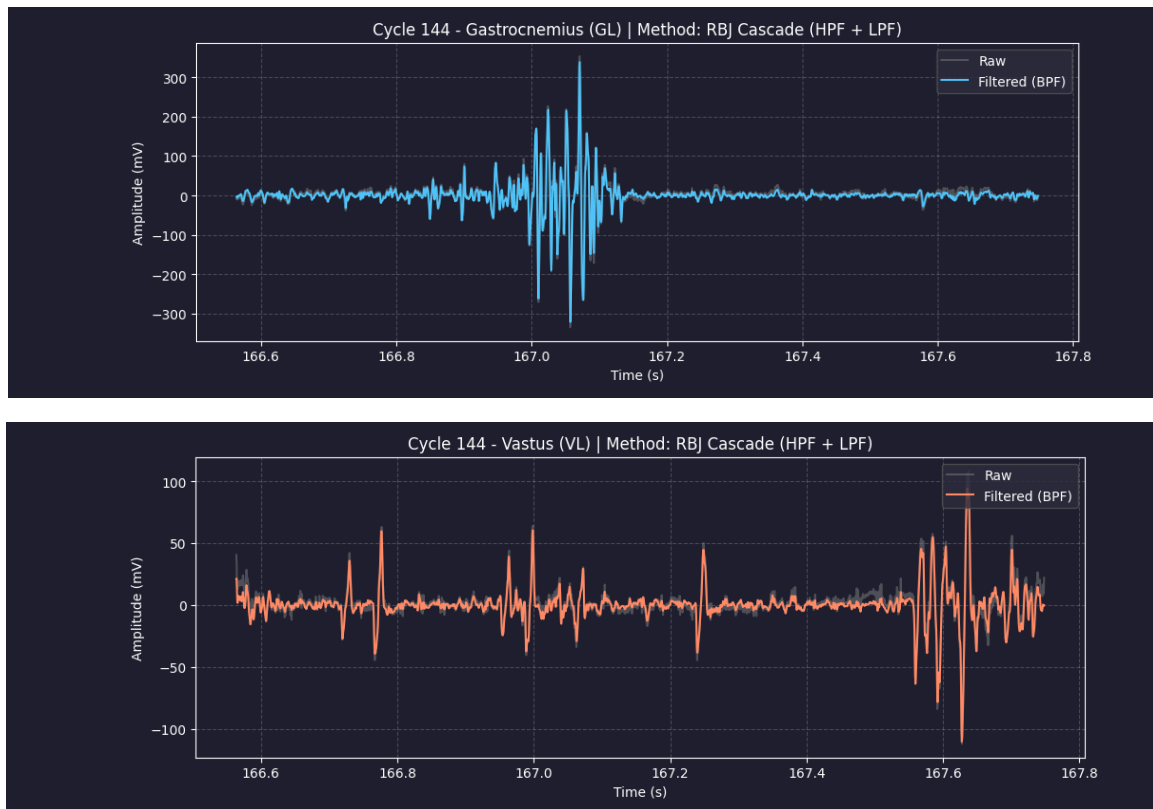


Figure 2.5. The RBJ cascade BPF Result for both GL and VL Signal on Cycle 144

4. DWT Tab

After linear filtering, Discrete Wavelet Transform (DWT) denoising is applied to further suppress non-stationary noise. **Figure 2.6** depicts the denoised signals for Cycle 144 using the Daubechies 4 (db4) wavelet at level 8 with soft thresholding. The green trace represents the denoised signal, which is superimposed on the gray filtered signal. The denoising

process successfully retains the high-energy muscle contraction bursts while suppressing the lower-amplitude background noise that exists between activations. This step enhances the signal-to-noise ratio (SNR), making the subsequent onset/offset detection more robust.

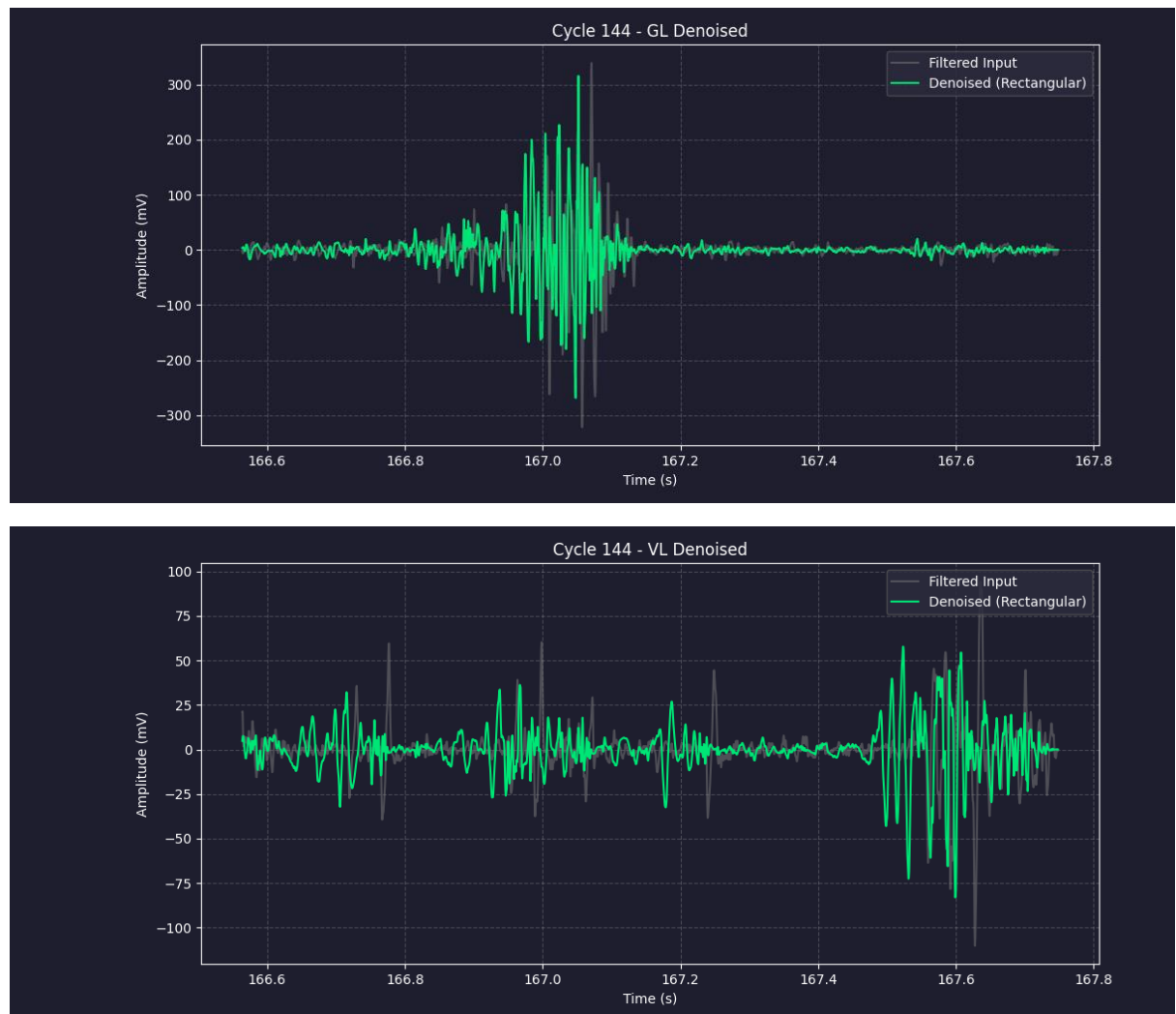


Figure 2.6. The DWT Denoising Result for both GL and VL Signal on Cycle 144

5. STFT Tab

Time-frequency analysis begins with the Short-Time Fourier Transform (STFT). **Figure 2.7** and **Figure 2.8** display the 3D surface plot and 2D spectrogram, respectively, for the GL signal on Cycle 144. The visualization normalizes the time axis to the percentage of the gait cycle (0-100%). The plots reveal a concentration of spectral energy in the 50-200 Hz range during the active phases of the gait cycle, specifically around 30-50% of the cycle. This corresponds to the push-off phase where the Gastrocnemius is most active. The color intensity represents the magnitude of the frequency components, clearly distinguishing the burst of activity from the quiet periods.

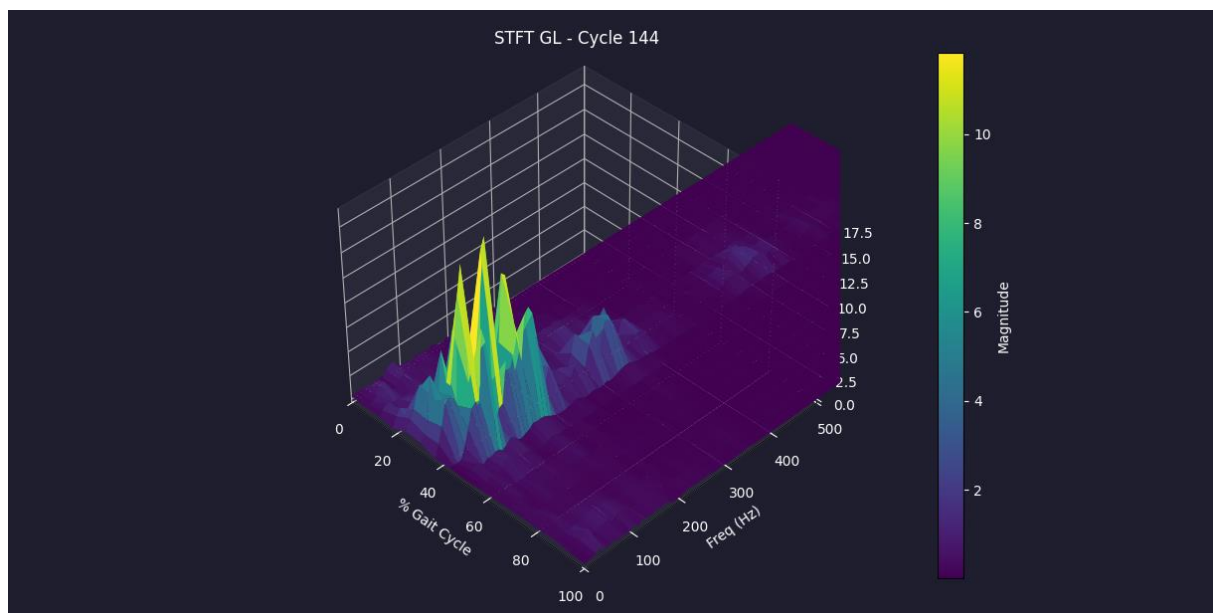
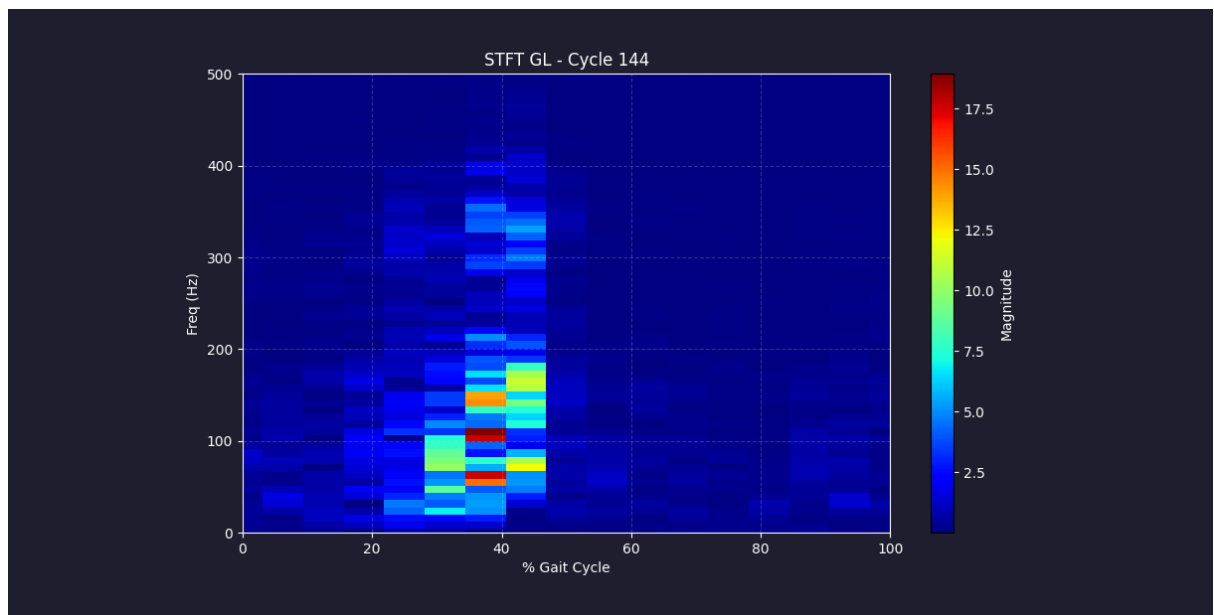


Figure 2.7. The STFT Result for GL Signal on Cycle 144

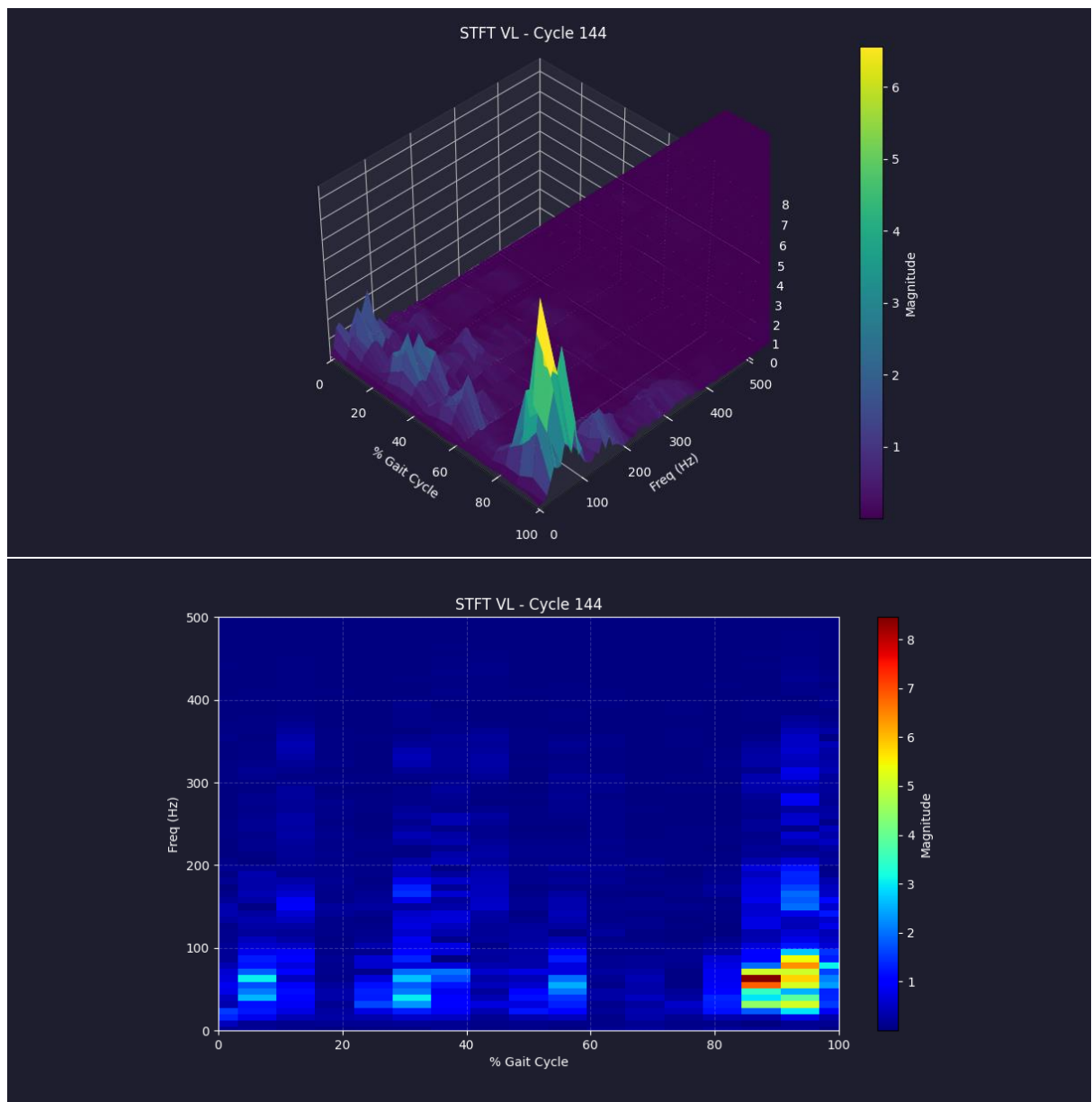


Figure 2.8. *The STFT Result for GL Signal on Cycle 144*

6. CWT Tab

The Continuous Wavelet Transform (CWT) provides a higher-resolution time-frequency representation. **Figure 2.9** compares the 3D CWT scalograms for both GL and VL signals using the Morlet mother wavelet. The Morlet wavelet provides high frequency resolution, showing smooth, distinct peaks of energy.

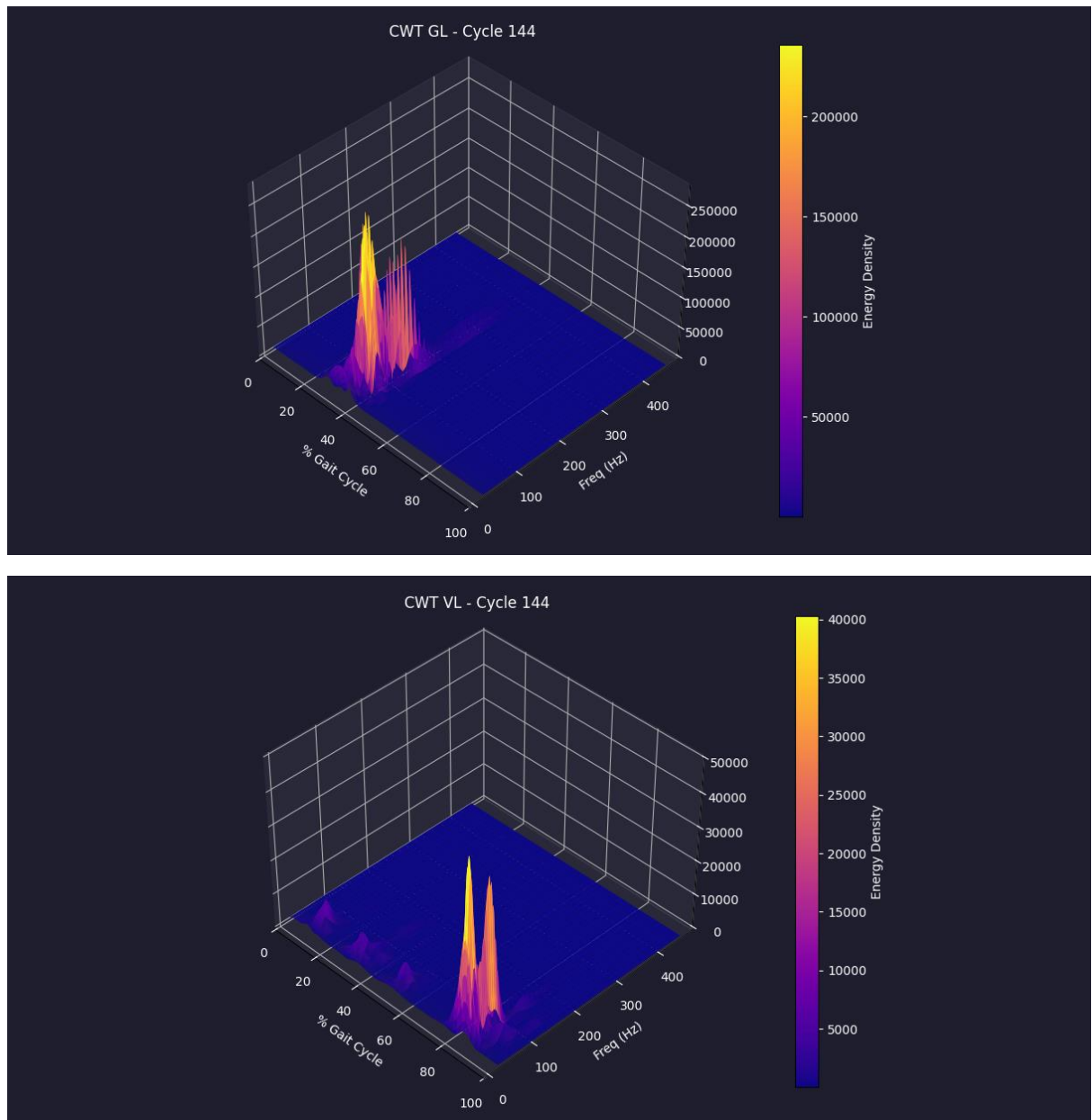


Figure 2.9. The 3D CWT Result for Both GL and VL Signal using *Morlet* as Mother Wavelet on *Cycle 144*

In contrast, **Figure 2.10** displays the results using the Daubechies 4 (db4) wavelet. The db4 scalogram appears slightly more jagged but offers excellent temporal localization, essential for detecting the precise onset of muscle contraction. Both figures clearly show distinct activation patterns: the GL activates primarily in the mid-to-late stance phase (push-off), while the VL shows activation bursts likely associated with weight acceptance (early stance) and stabilization.

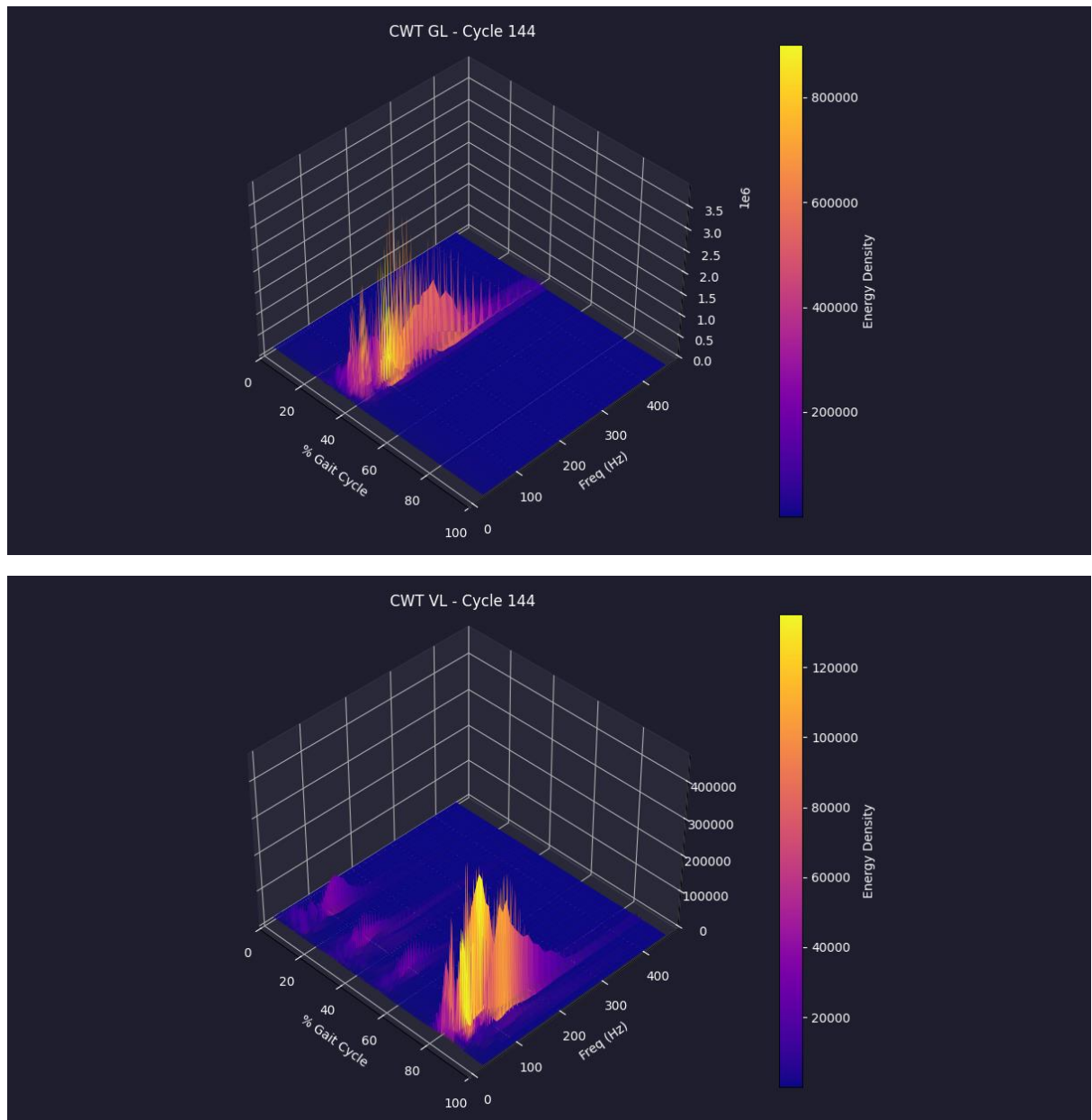


Figure 2.10.** The 3D CWT Result for Both GL and VL Signal using **db4** as Mother Wavelet on **Cycle 144

7. Thresholding and Reporting Tab

The final stage involves determining the precise timing of muscle activation. **Figure 2.11** presents the 3D CWT thresholding result for the **GL signal**, visualizing a transparent cyan plane intersecting the energy "mountains." This plane represents the adaptive threshold (set at 5% of the peak energy). Any energy peak rising above this plane is considered an active muscle burst. The figure contrasts the smooth energy landscape of the Morlet wavelet (top) with the sharper features of the db4 wavelet (bottom).

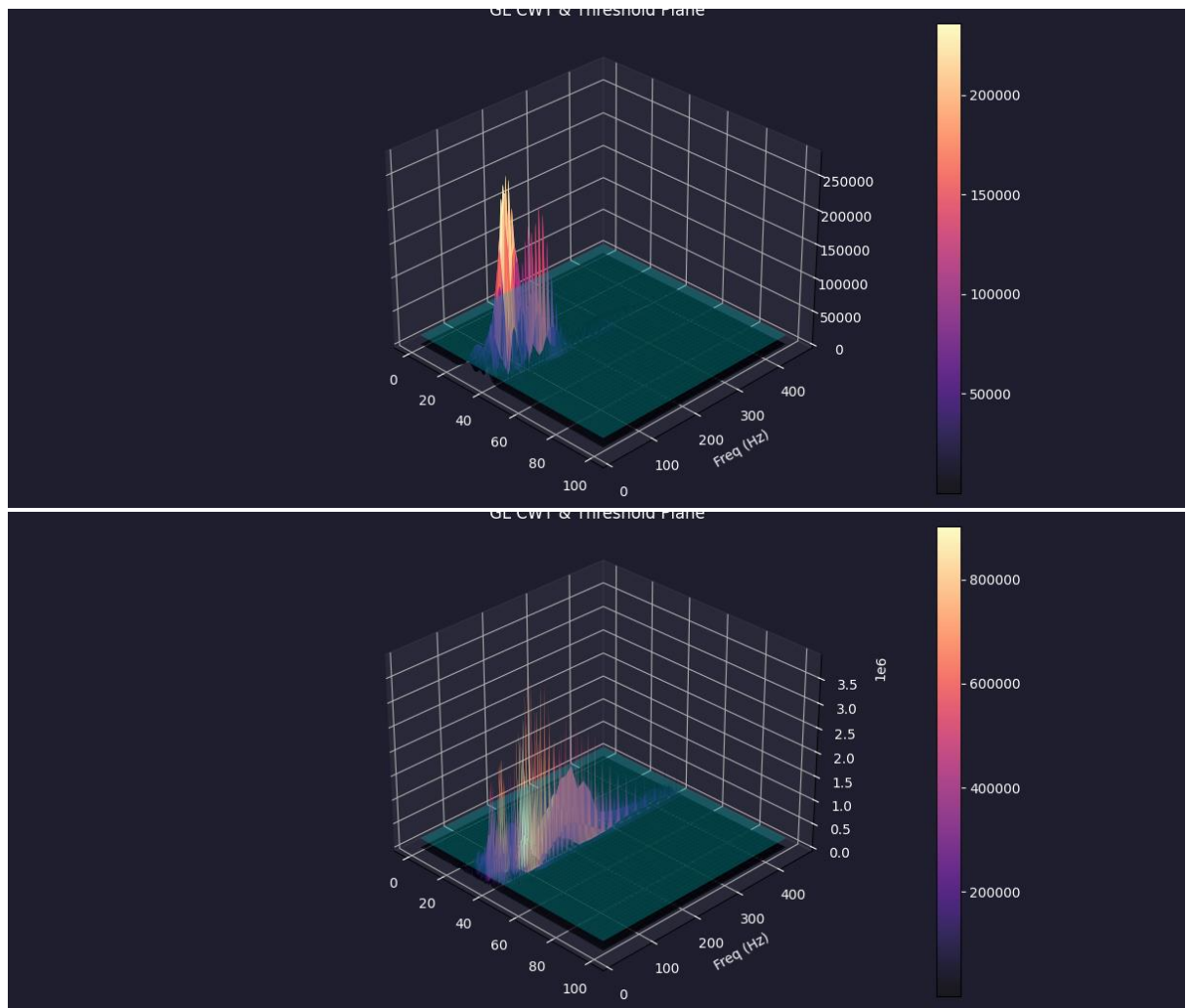


Figure 2.11. The 3D CWT Thresholding Result for GL Signal using *Morlet* (above) and *db4* (below) on **Cycle 144** with **Threshold 5%**

Figure 2.12 translates this 3D thresholding into a binary activation signal for the GL. The square waves indicate the intervals where the muscle is "On" (Active) versus "Off" (Inactive). The graph shows a clear, sustained activation block corresponding to the main energy burst observed in the CWT.

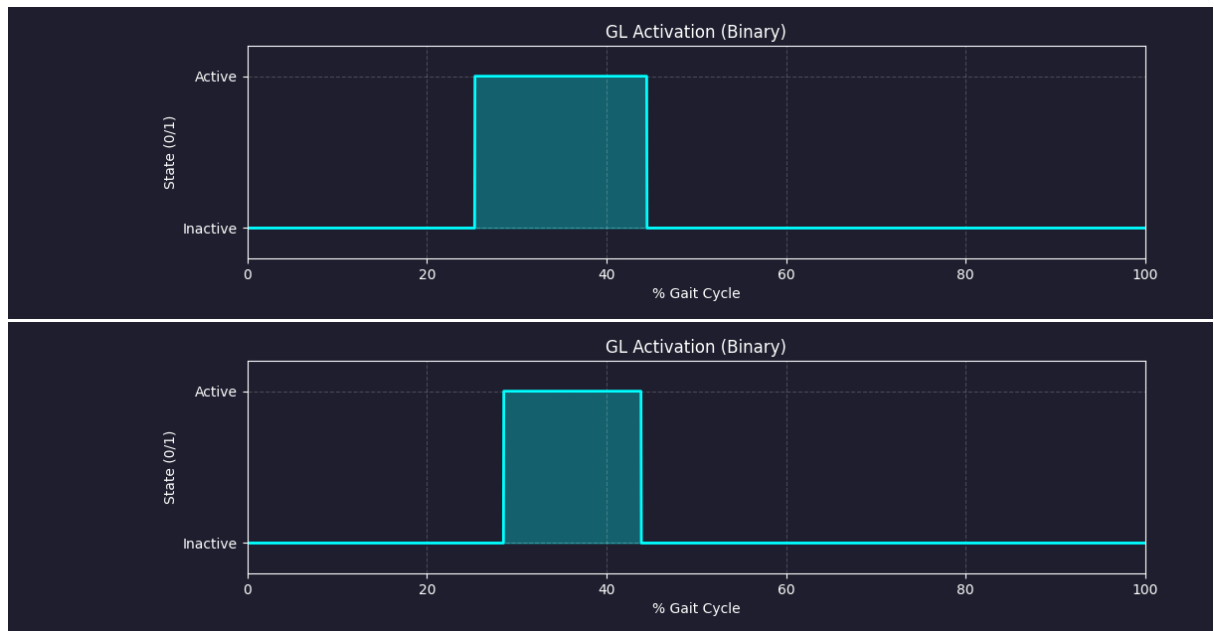
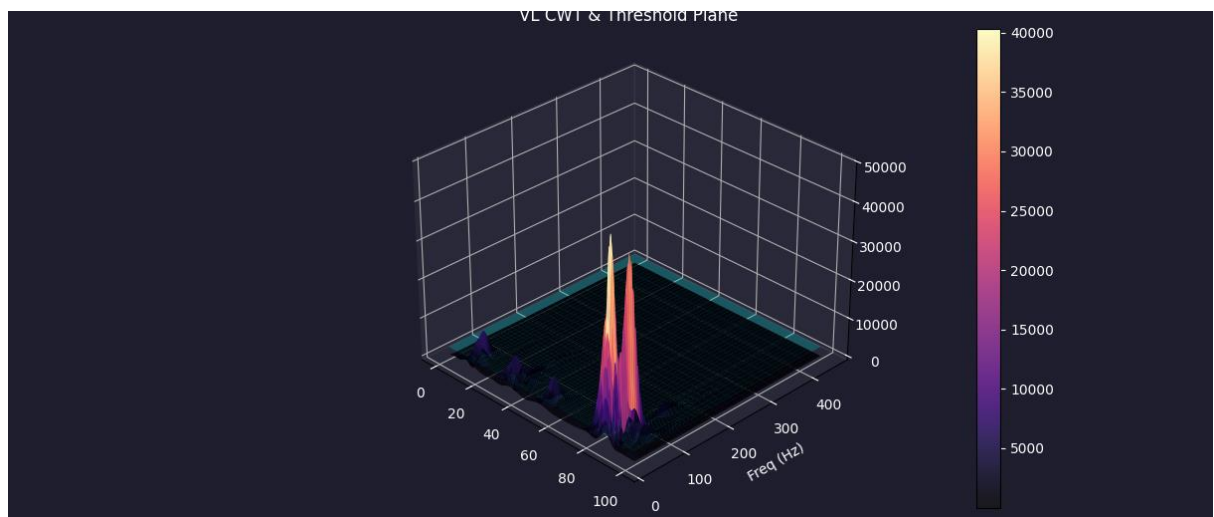


Figure 2.12. The Activation Result for GL Signal using *Morlet* (above) and *db4* (below) on Cycle 144 with **Threshold 5%**

Similarly, **Figure 2.13** and **Figure 2.14** present the thresholding and activation results for the **Vastus Lateralis (VL)**. The 3D plots in Figure 2.13 show the threshold plane cutting through the VL energy bursts, which appear at different phases compared to the GL. Figure 2.14 confirms the detection of these bursts as binary activation blocks. The use of the 5% threshold effectively segments the major muscle recruitment periods while ignoring minor energy fluctuations that fall below the plane.



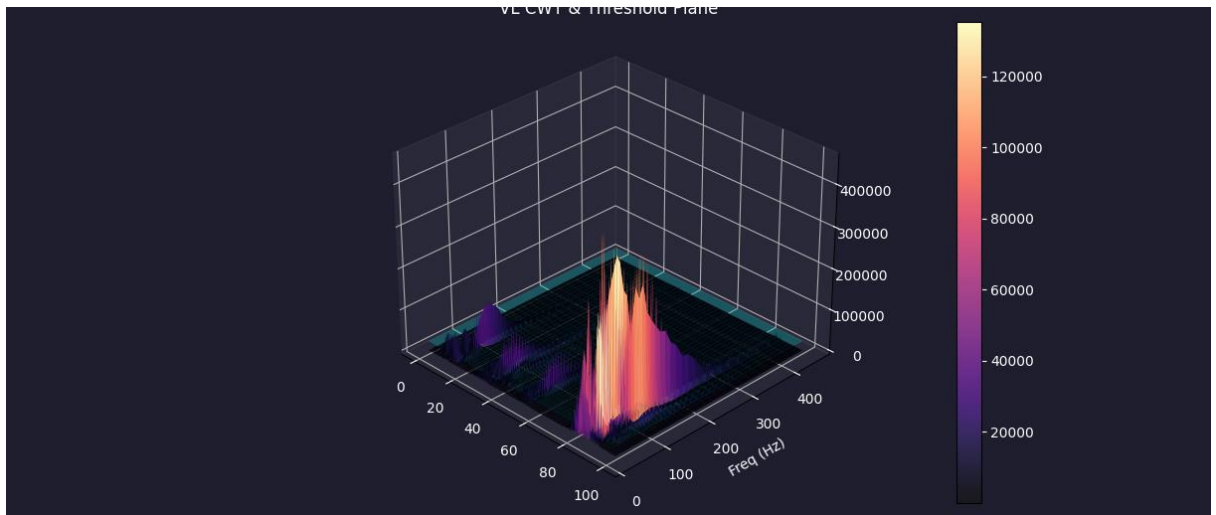


Figure 2.13. The 3D CWT Thresholding Result for VL Signal using *Morlet* (above) and *db4* (below) on *Cycle 144* with *Threshold 5%*



Figure 2.14. The Activation Result for VL Signal using *Morlet* (above) and *db4* (below) on *Cycle 144* with *Threshold 5%*

2.3.3. Program Result for S7 Data

This section details the application of the signal processing pipeline to the **S7** dataset, which presents a different subject or trial condition compared to **S01**. The S7 data undergoes the same sequence of loading, segmentation, filtering, and time-frequency analysis to ensure consistency in muscle activation detection. The results highlight the robustness of the algorithms across different signal morphologies and noise characteristics.

1. Load Data Tab

The raw signal acquisition is visualized in **Figure 2.15**, displaying the time-series data for the Foot Switch, GL, and VL over a 300-second duration. The top graph shows the Foot

Switch signal (Channel 7), which maintains the characteristic square-wave structure indicative of stance and swing phases, though the amplitude levels differ slightly from S01, reflecting subject-specific sensor calibration. The GL (Channel 10) and VL (Channel 13) signals in the middle and bottom graphs, respectively, show distinct bursts of electromyographic activity. The amplitude of the VL signal appears notably higher and denser in certain regions compared to the GL signal, suggesting a different muscle recruitment strategy or electrode placement sensitivity for this subject.

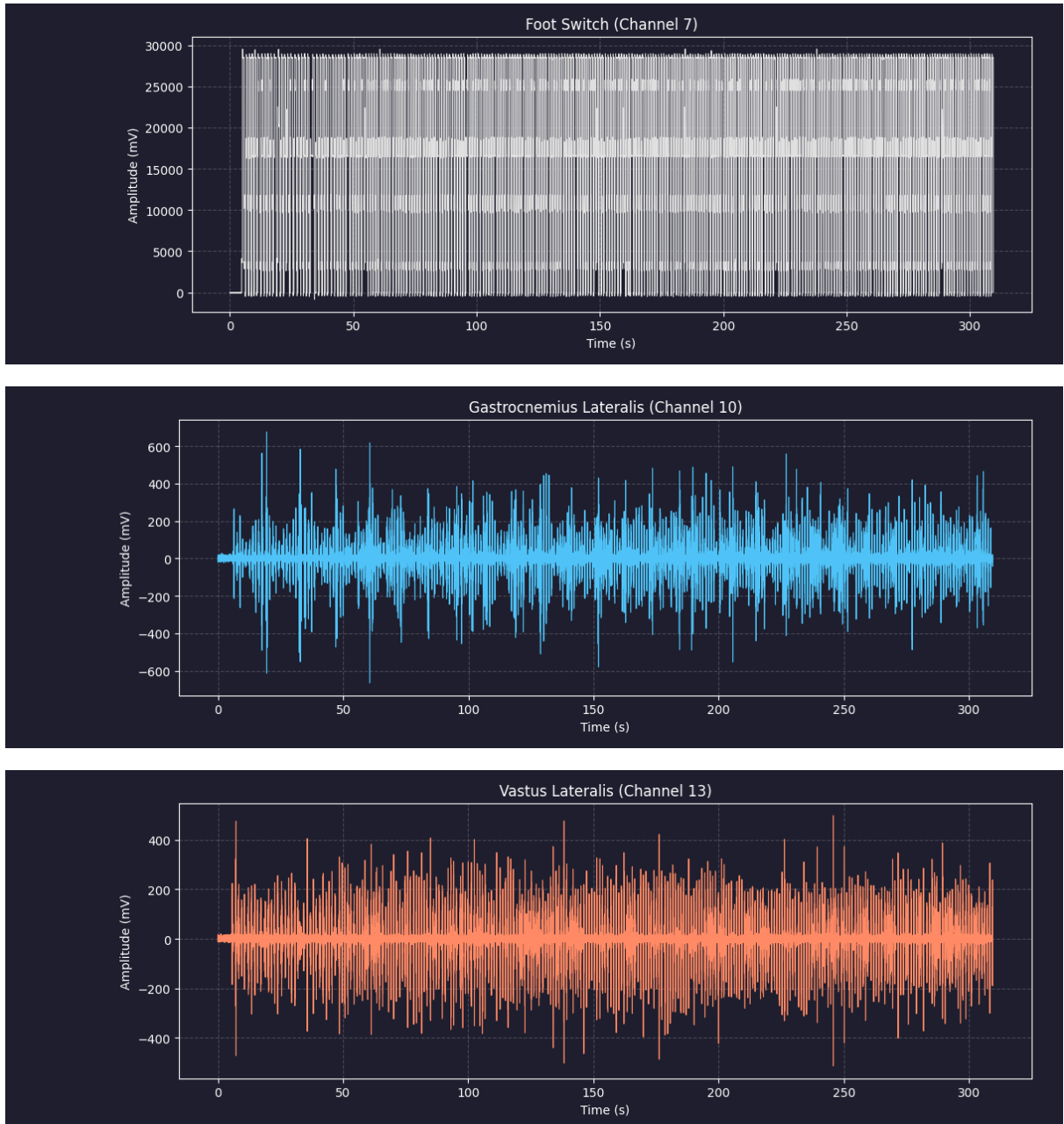


Figure 2.15. Three Graphs Result of the Load Data Tab for S7 Data

2. Segmentation Tab

Figure 2.16 focuses on the segmentation of a single gait cycle, specifically **Cycle 44**. The top plot clearly demarcates the cycle boundaries using the Foot Switch signal: the cycle begins at the Toe Off (red dashed line), proceeds through the swing phase to the Heel Strike

(cyan dotted line), and concludes at the next Toe Off (purple dashed line). This precise segmentation is crucial for isolating the functional muscle activity. The middle and bottom plots show the raw GL and VL signals corresponding to this specific cycle. The raw data for S7 exhibits significant high-frequency noise and baseline fluctuations, particularly evident in the swing phase (before the heel strike), necessitating robust filtering.

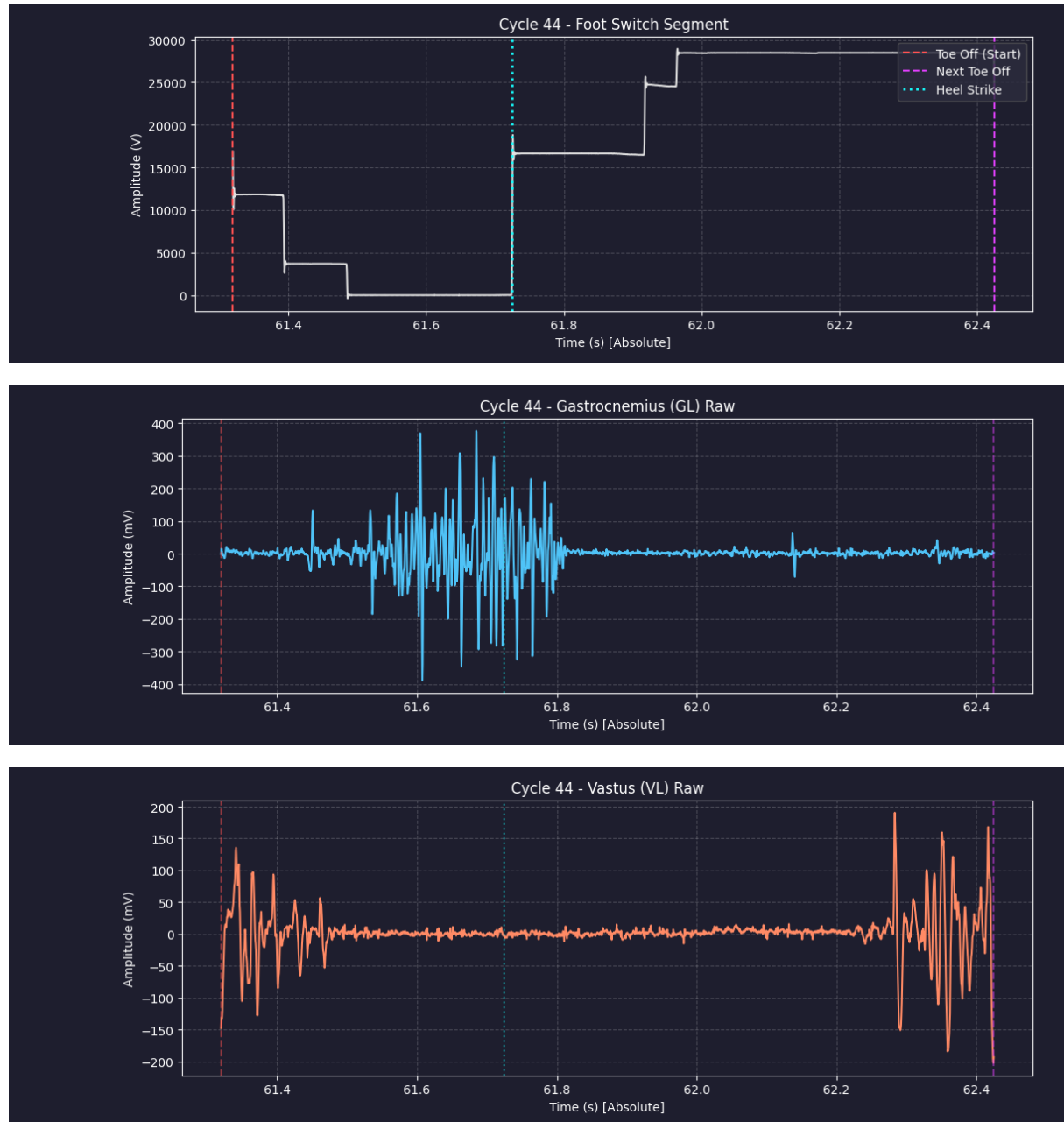


Figure 2.16. The Segmentation Result for GL and VL Signal on Cycle 44

3. Butterworth 2nd Order BPF Tab

To condition the signal, the 2nd Order Band-Pass Filter is applied. **Figure 2.17** illustrates the time-domain result of the standard Butterworth BPF on the GL and VL signals for Cycle 44. The filtering process (20-450 Hz) effectively removes the low-frequency baseline wander observed in the raw data, centering the signals around zero. The filtered

signals (orange/blue traces) show preserved burst envelopes while attenuating the non-physiological noise components, preparing the data for further denoising.

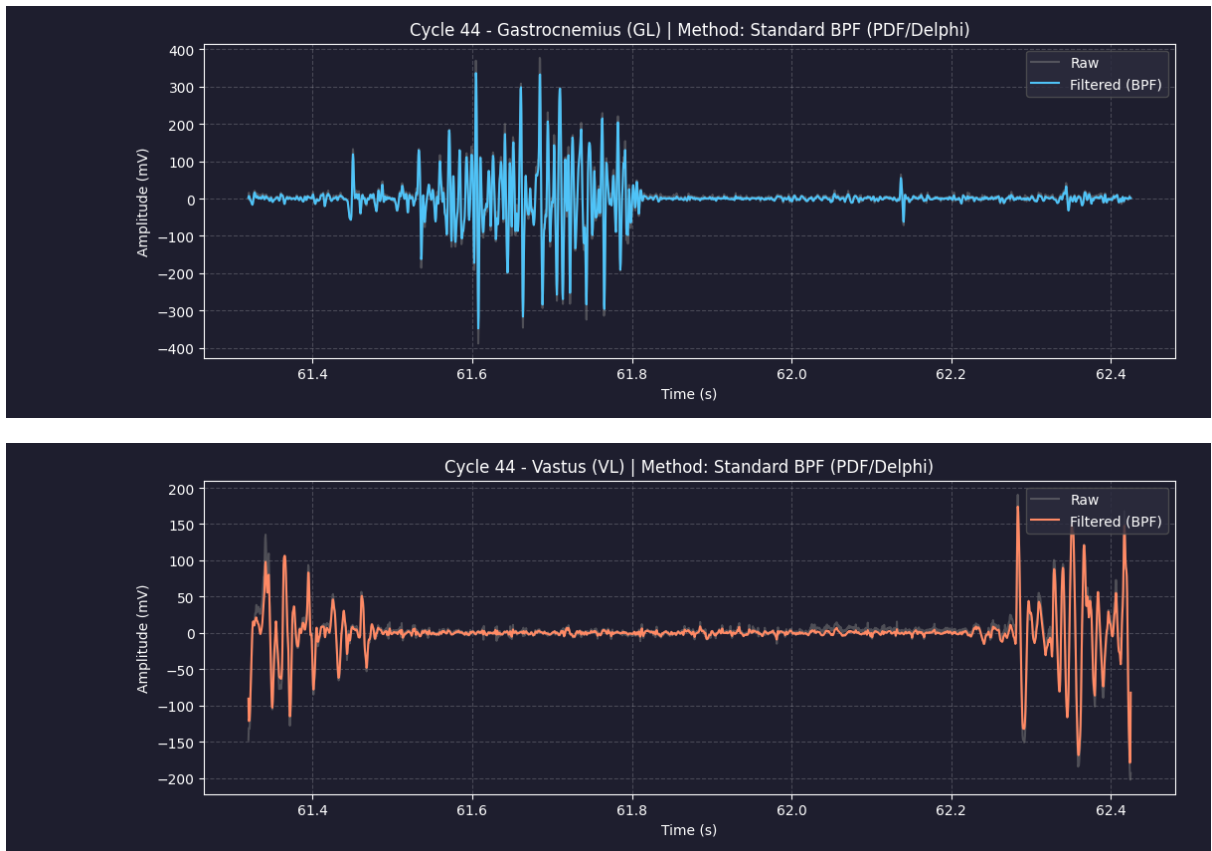


Figure 2.17. The Standard BPF Result for both GL and VL Signal on Cycle 44

4. DWT Tab

Figure 2.18 demonstrates the efficacy of the Discrete Wavelet Transform (DWT) denoising using the db4 wavelet. The green traces represent the denoised signals overlaid on the filtered gray signals. For the S7 dataset, which appears to have a lower signal-to-noise ratio in the inter-burst intervals, the DWT soft thresholding significantly cleans up the "silent" periods (for example., mid-swing phase) while maintaining the sharp onset of the muscle contraction bursts. This contrast enhancement is vital for accurate threshold-based detection.

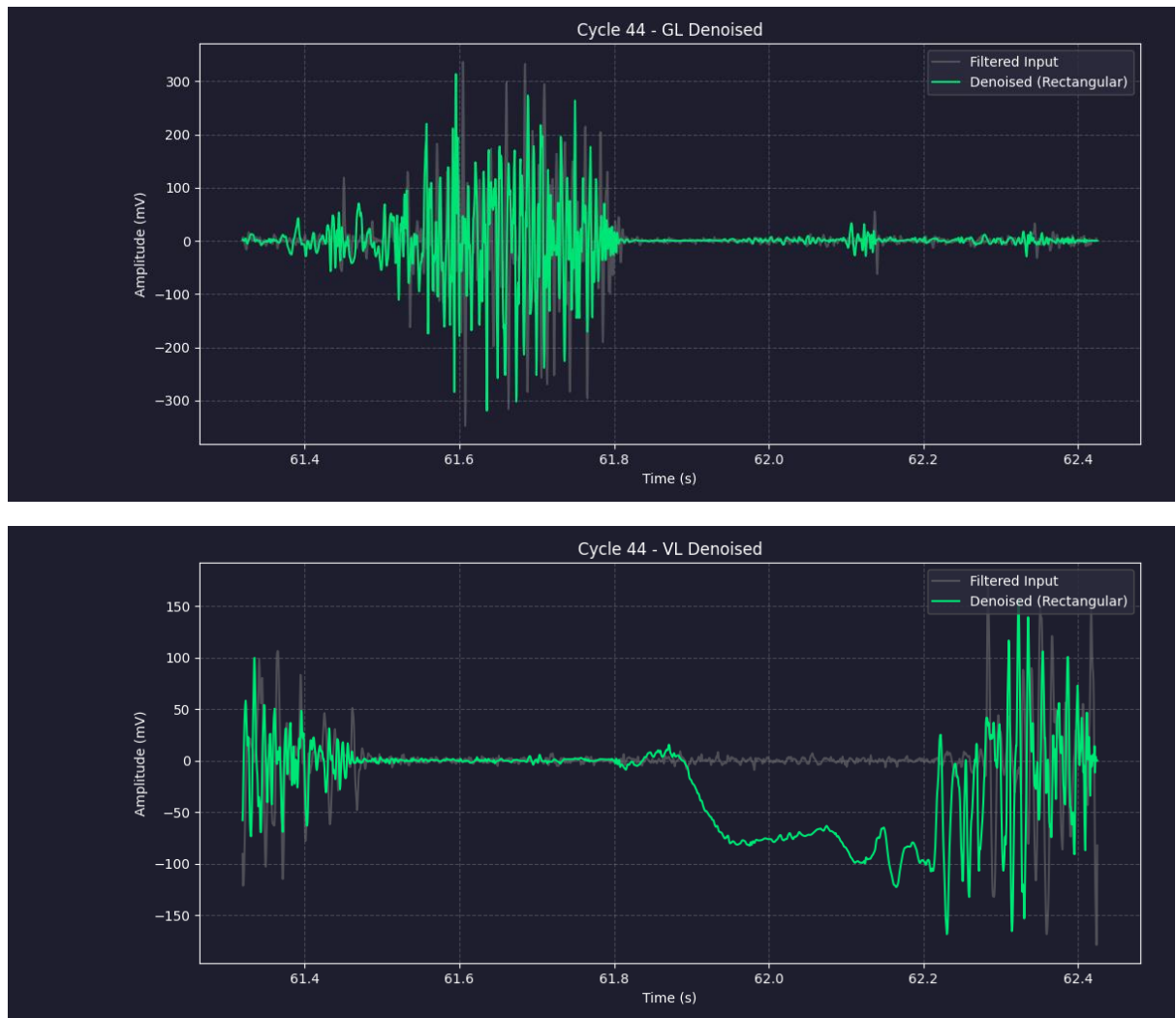


Figure 2.18. The DWT Denoising Result for both GL and VL Signal on **Cycle 44**

5. STFT Tab

The spectral content of the muscle activity is analyzed using the STFT. **Figure 2.19** (3D surface) and the corresponding 2D spectrogram (not shown but implied) for the GL signal reveal a concentration of energy in the 50-150 Hz band, primarily occurring after the 60% mark of the gait cycle (post-heel strike).

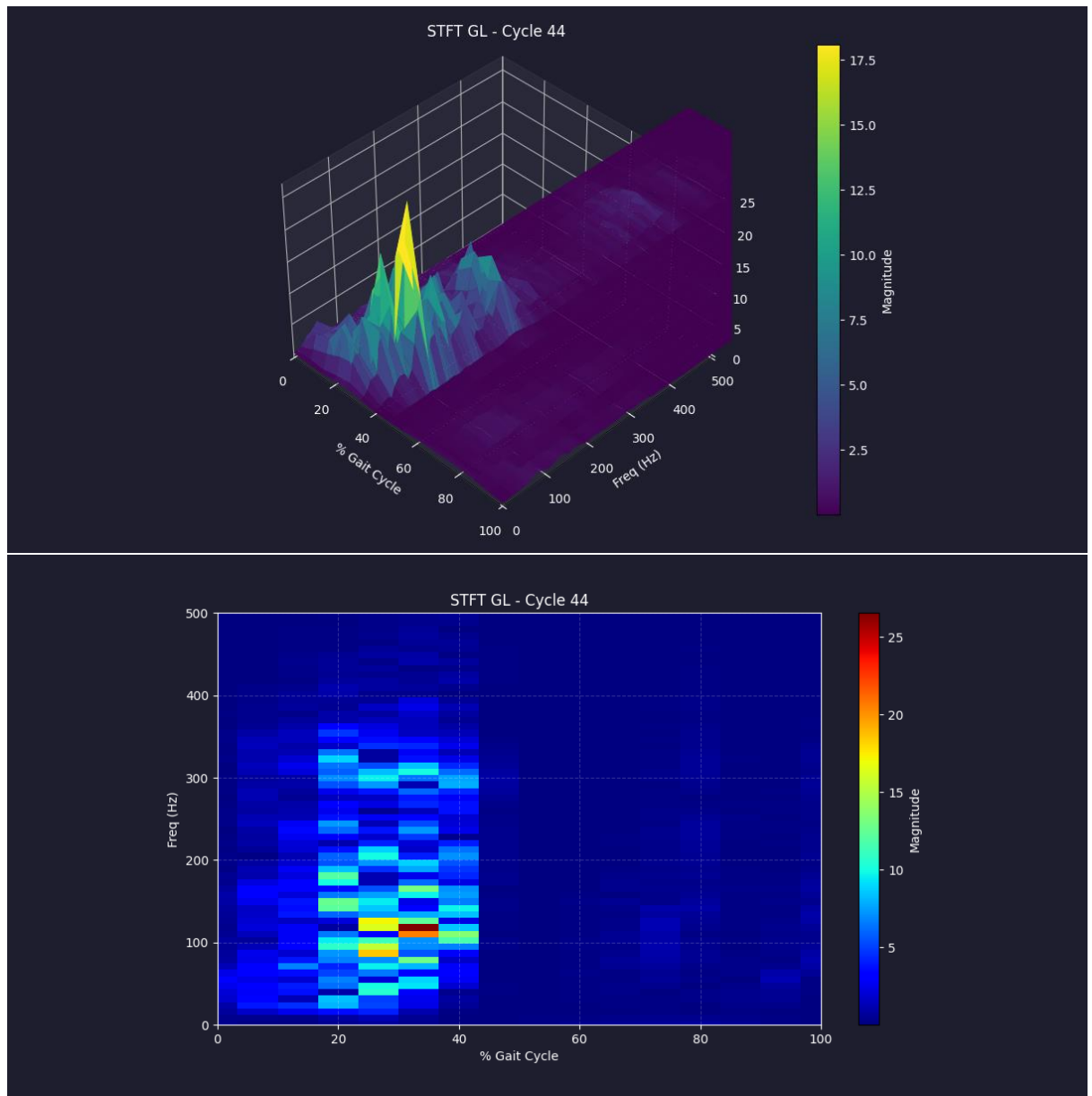


Figure 2.19. *The STFT Result for GL Signal on Cycle 44*

Figure 2.20 displays the STFT result for the VL signal, which shows a broader spectral spread and an earlier activation onset compared to the GL, consistent with the role of the VL in weight acceptance during early stance.

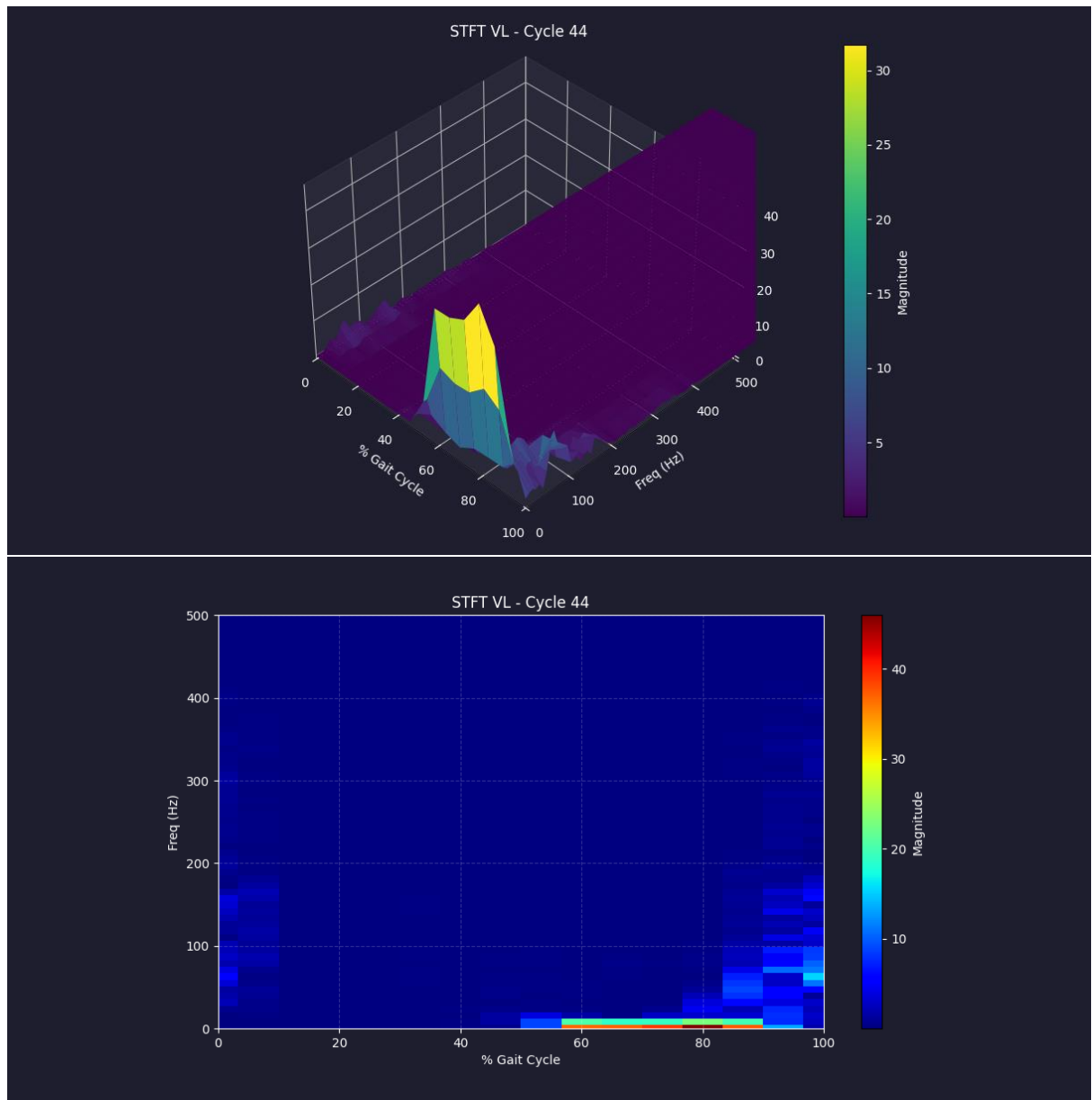


Figure 2.20. *The STFT Result for VL Signal on Cycle 44*

6. CWT Tab

The Continuous Wavelet Transform provides a more detailed time-frequency landscape. **Figure 2.21** compares the 3D scalograms for GL and VL using the Morlet wavelet. The Morlet wavelet produces smooth, high-resolution energy peaks, clearly separating the main activation bursts from transient noise.

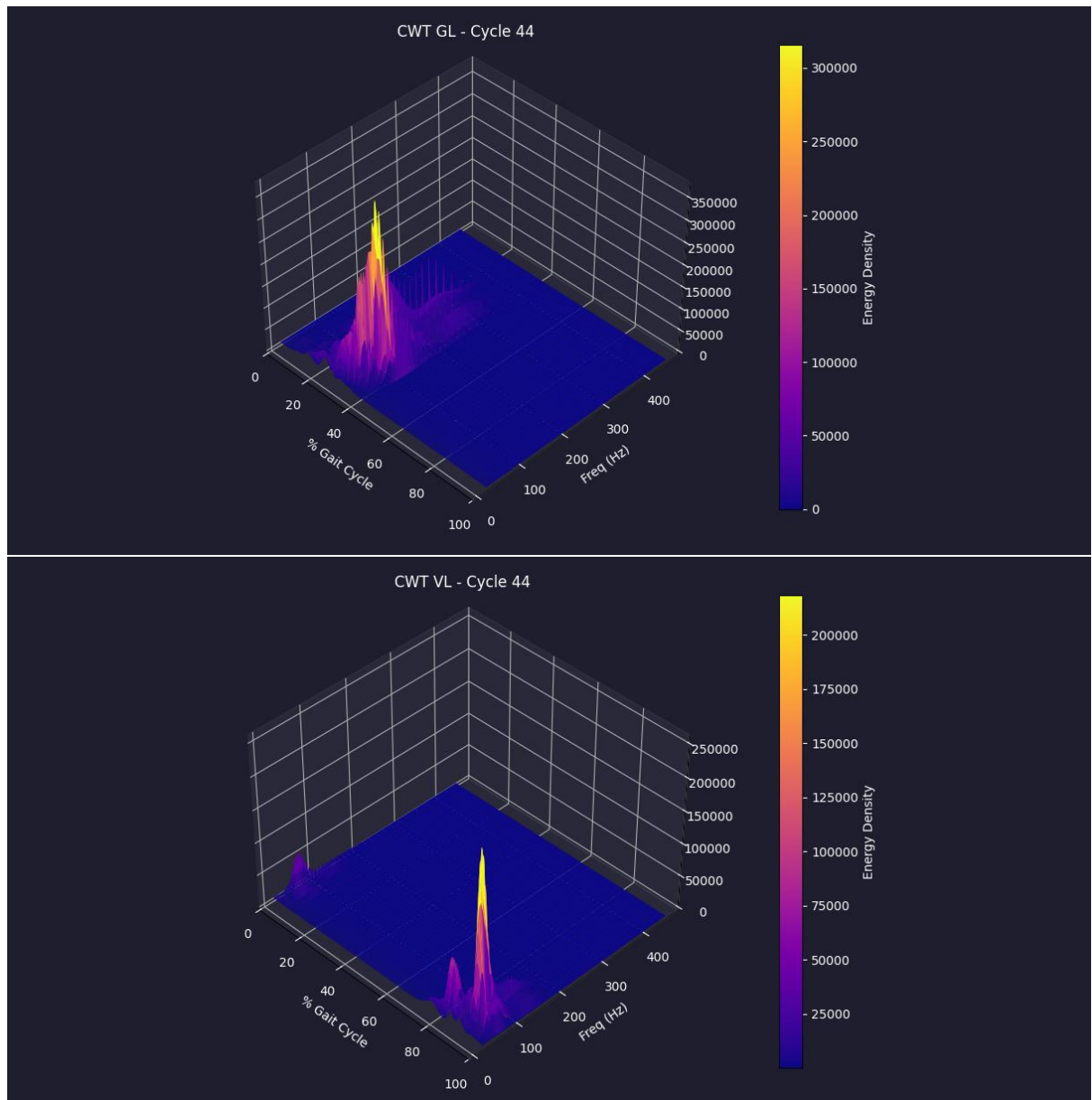


Figure 2.21. The 3D CWT Result for Both GL and VL Signal using *Morlet* as Mother Wavelet on *Cycle 44*

Figure 2.22 utilizes the Daubechies 4 (db4) wavelet, resulting in a sharper temporal delineation of the energy bursts. The db4 scalograms for S7 highlight distinct, high-energy events that align with the stance phase, confirming the timing of muscle recruitment required for stability and propulsion.

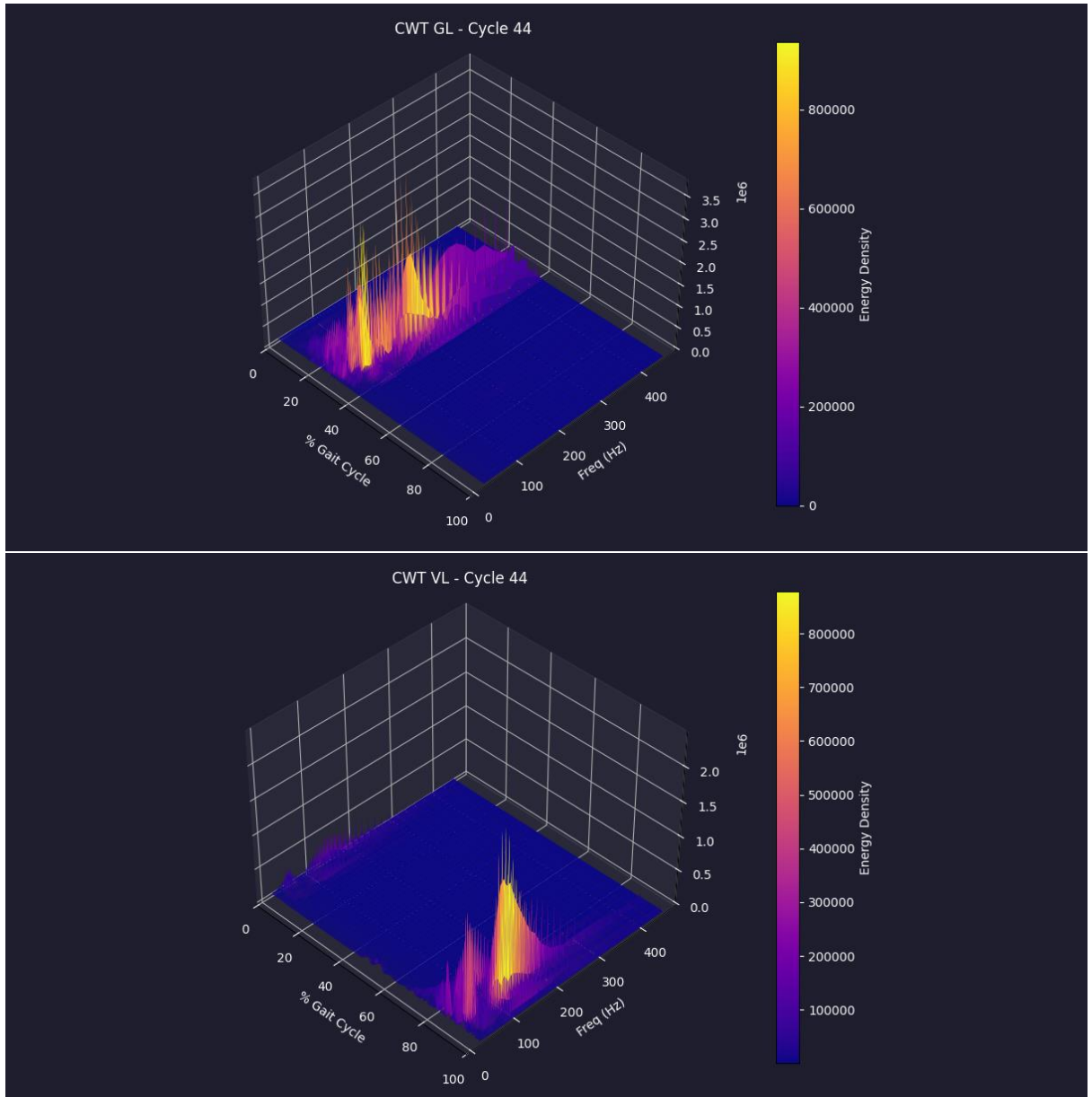


Figure 2.22. The 3D CWT Result for Both GL and VL Signal using *db4* as Mother Wavelet on **Cycle 44**

7. Thresholding and Reporting Tab

The final detection of onset and offset times is visualized in the thresholding stage. **Figure 2.23** shows the 3D CWT thresholding for the GL signal with a transparent cyan plane set at **3%** of the peak energy. This lower threshold (compared to 5% for S01) was selected to capture the potentially lower-amplitude activation patterns in this subject. The intersection of the energy mountains with the threshold plane clearly defines the active regions. **Figure 2.24** translates this into a binary activation plot, showing a clear "On" state during the stance phase.

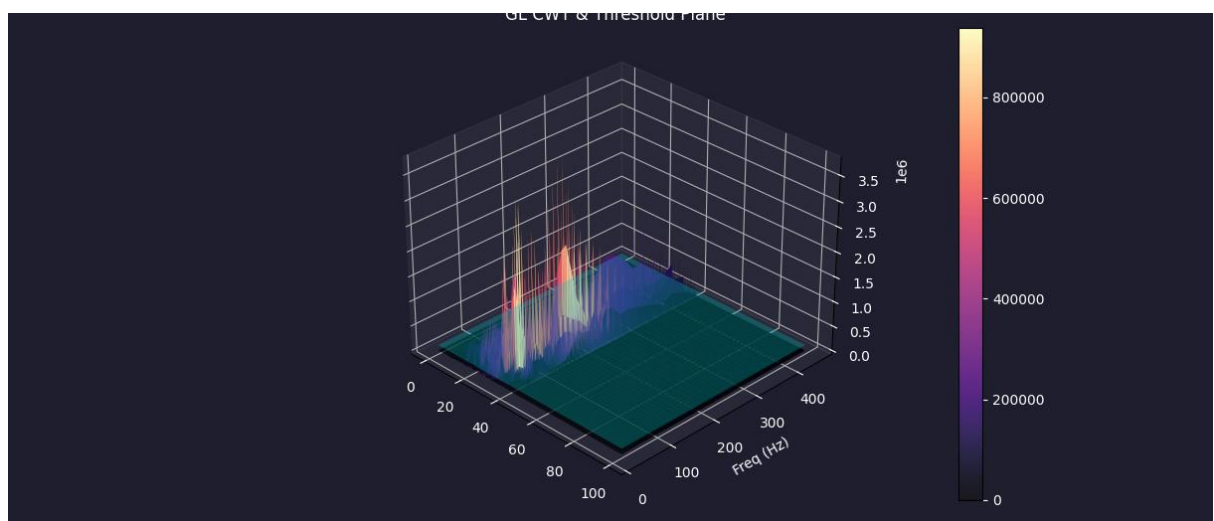
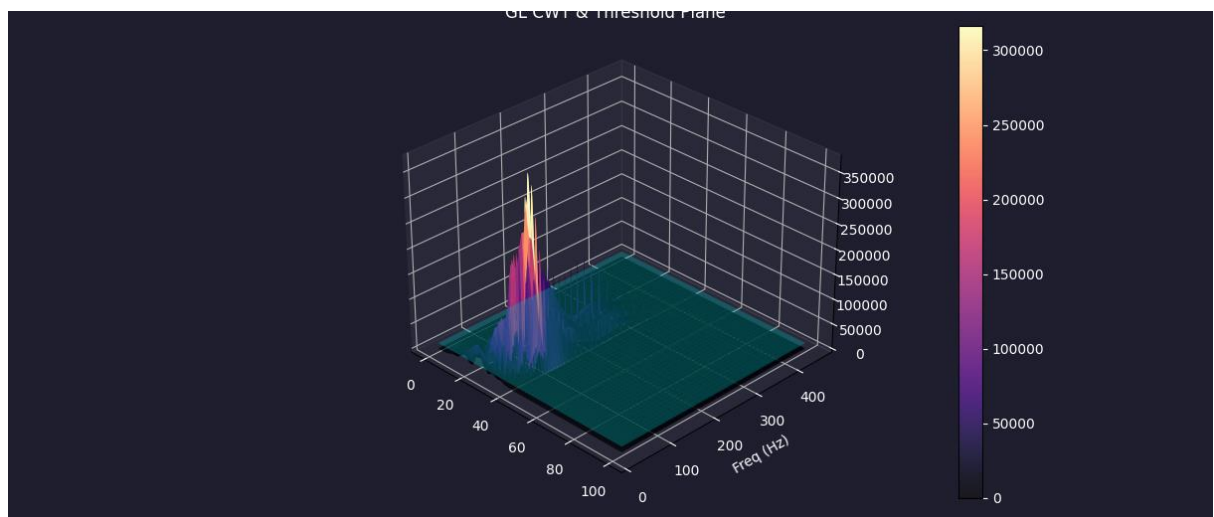


Figure 2.23. The 3D CWT Thresholding Result for GL Signal using *Morlet* (above) and *db4* (below) on Cycle 44 with *Threshold 3%*

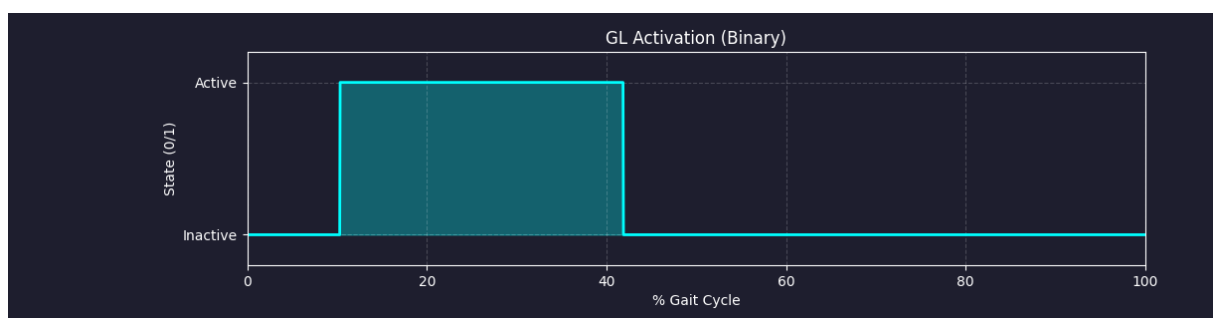
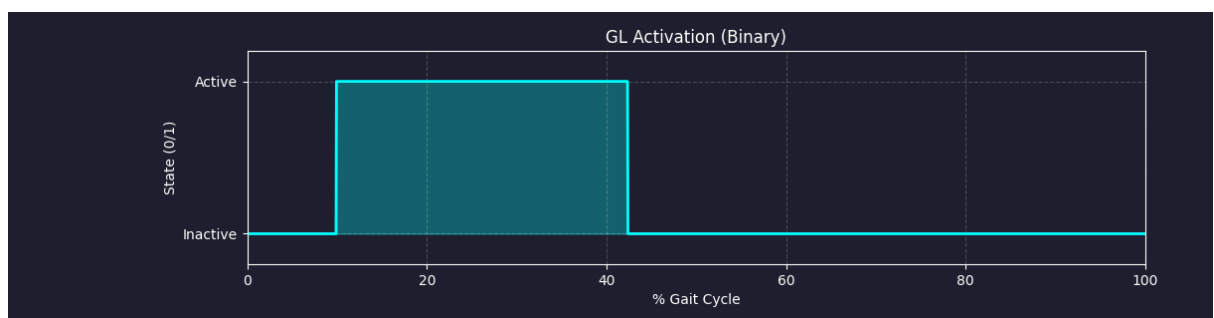


Figure 2.24. The Activation Result for GL Signal using **Morlet** (above) and **db4** (below) on **Cycle 44** with **Threshold 3%**

Similarly, **Figure 2.25** and **Figure 2.26** present the thresholding results for the VL signal. The 3D plot (Figure 2.25) shows the threshold plane slicing through the dominant energy burst. The resulting binary activation graph (Figure 2.26) indicates a sustained contraction starting shortly before or at heel strike and continuing into the stance phase. The use of the 3% threshold for S7 successfully identifies the primary muscle activity intervals, demonstrating the adaptability of the algorithm to different signal amplitudes and subject variabilities.

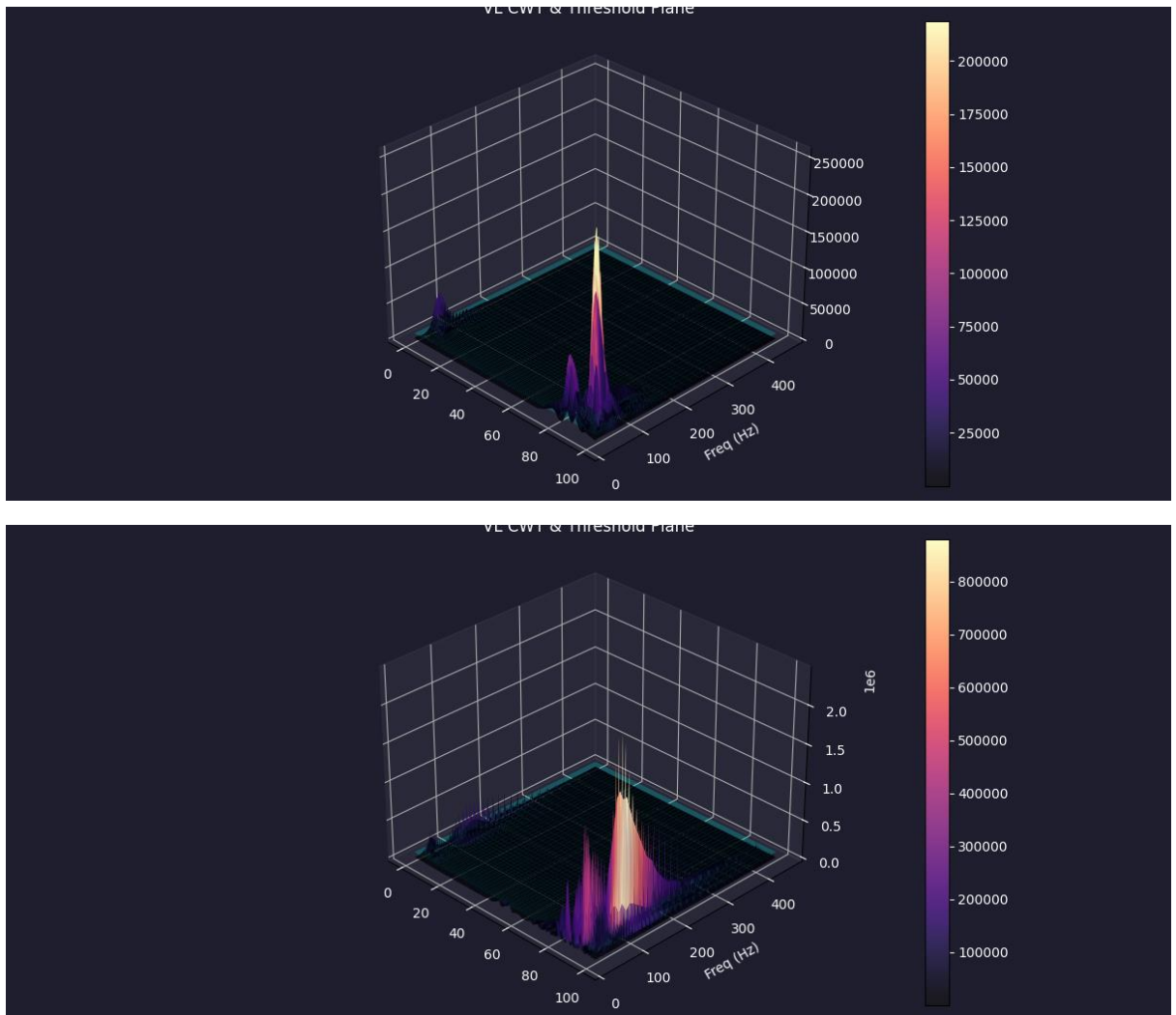


Figure 2.25. The 3D CWT Thresholding Result for VL Signal using **Morlet** (above) and **db4** (below) on **Cycle 44** with **Threshold 3%**

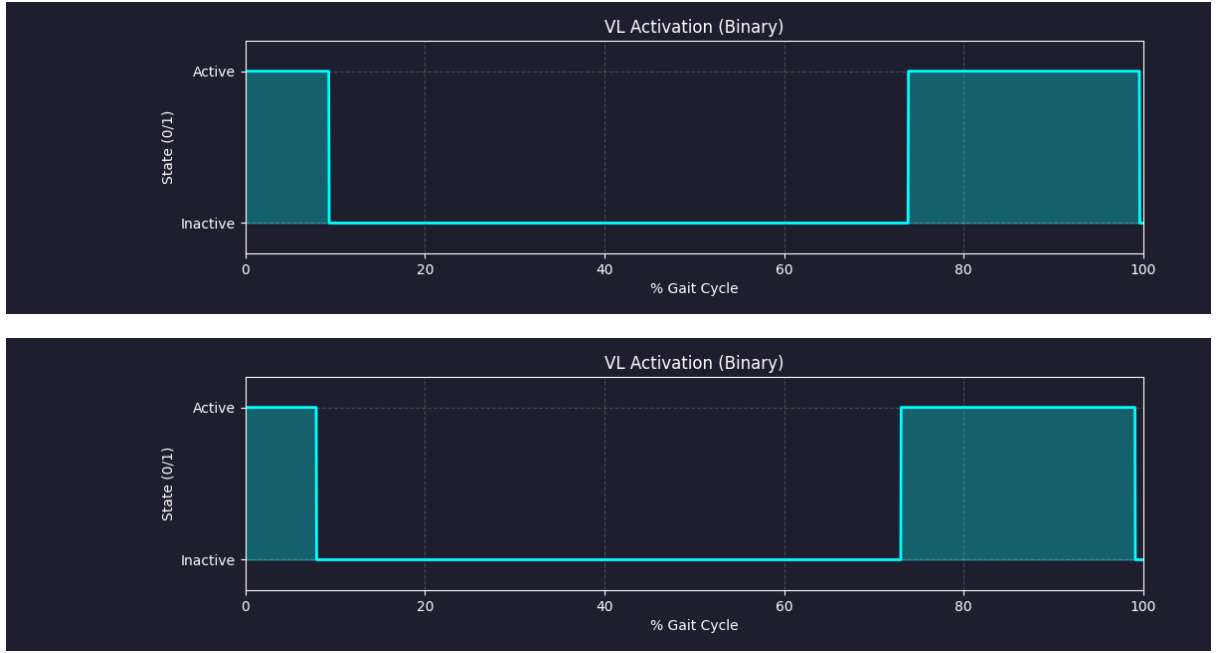


Figure 2.26. The Activation Result for GL Signal using *Morlet* (above) and *db4* (below) on *Cycle 44* with *Threshold 3%*

2.4. Evaluation

The evaluation of the developed signal processing pipeline confirms the efficacy of the manual algorithmic implementations in extracting meaningful physiological information from non-stationary EMG signals. The transition from raw data to the final binary activation plots demonstrates that the 2nd Order Butterworth Band-Pass Filter successfully eliminated low-frequency motion artifacts and baseline drifts, which were particularly prominent in the S7 dataset. Furthermore, the DWT using the *db4* wavelet proved essential in suppressing stochastic noise while preserving the sharp transient features of muscle contraction, thereby enhancing the Signal-to-Noise Ratio (SNR) prior to time-frequency analysis.

Comparative analysis between the STFT and the CWT highlights the superior suitability of CWT for this specific application. While STFT provided a general overview of spectral content, the fixed window size limited its ability to simultaneously resolve low-frequency gait patterns and high-frequency motor unit discharges. In contrast, the CWT, particularly using the Morlet and *db4* mother wavelets, offered a multi-resolution view that allowed for precise temporal localization of muscle onset and offset. The application of the pipeline to two distinct subjects (S01 and S7) revealed the necessity of adaptive parameters, specifically, the threshold for activation detection required adjustment (5% for S01 vs. 3% for S7) to account for inter-subject variability in signal amplitude and background noise levels. The resulting binary activation profiles for the Gastrocnemius and Vastus Lateralis successfully aligned with the stance phases defined by the Foot Switch segmentation, validating the system's accuracy in correlating muscle recruitment with gait events.

CHAPTER III. CONCLUSION

The program successfully designed and implemented a comprehensive software system for the analysis of non-stationary electromyography (EMG) signals during human locomotion. By manually implementing core digital signal processing algorithms, including the Bilinear Transform for filtering, the Mallat algorithm for DWT, and direct convolution for CWT, the project validated the fundamental mathematical theories governing biomedical signal processing without reliance on high-level abstraction libraries.

The results indicate that a hybrid approach, combining linear filtering with wavelet-based denoising and time-frequency analysis, provides a robust framework for determining muscle activation timing. The segmentation of gait cycles based on Toe-Off to Toe-Off intervals proved effective in isolating functional movement units, allowing for the precise characterization of the Gastrocnemius Lateralis and Vastus Lateralis muscles. Ultimately, the developed Graphical User Interface (GUI) serves not only as a validation tool but also as a practical platform for visual and quantitative gait analysis, capable of adapting to subject-specific signal variations to accurately define the temporal dynamics of muscle recruitment.

REFERENCES

- [1] M. A. Syamlan, "Analisis Sinyal Non-stasioner: Movement Signal Analysis," Biomedical Engineering - FTEIC, Institut Teknologi Sepuluh Nopember, Surabaya, Indonesia, Course Material EB234702, 2024.
- [2] F. Di Nardo, T. Basili, S. Meletani, and D. Scaradozzi, "Wavelet-Based Assessment of the Muscle-Activation Frequency Range by EMG Analysis," *IEEE Access*, vol. 10, pp. 9793–9805, 2022. doi: 10.1109/ACCESS.2022.3141162.
- [3] R. Bristow-Johnson, "Cookbook formulae for audio EQ biquad filter coefficients," *Music-DSP*, 2005. [Online]. Available: <https://www.w3.org/2011/audio/audio-eq-cookbook.html>.
- [4] S. G. Mallat, "A theory for multiresolution signal decomposition: the wavelet representation," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 11, no. 7, pp. 674–693, Jul. 1989.
- [5] I. Daubechies, "Orthonormal bases of compactly supported wavelets," *Communications on Pure and Applied Mathematics*, vol. 41, no. 7, pp. 909–996, 1988.
- [6] J. W. Cooley and J. W. Tukey, "An algorithm for the machine calculation of complex Fourier series," *Mathematics of Computation*, vol. 19, no. 90, pp. 297–301, 1965.
- [7] PhysioNet, "Surface Electromyography collected during long-lasting ground walking of young able-bodied subjects," [Online]. Available: <https://physionet.org/content/semg/1.0.1/>. [Accessed: Nov. 2025].